

Fundamentals: Markov Decision Process

**Kianté Brantley
CS 1840 Fall 2026**

Course Overview

Instructor: Kianté Brantley

TFs: Alex Cai, Philip Nielsen, Egil Rolstad

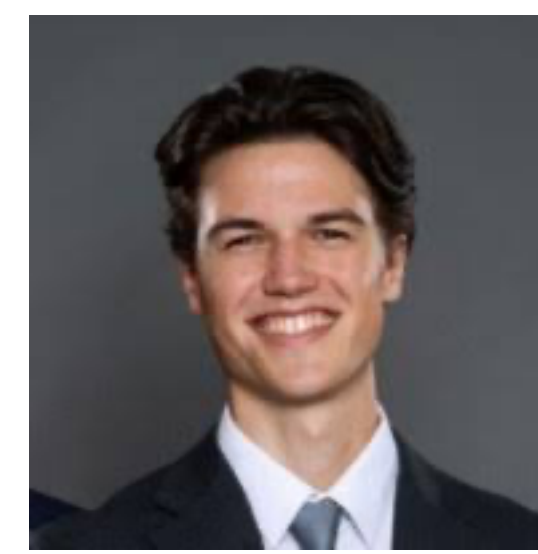
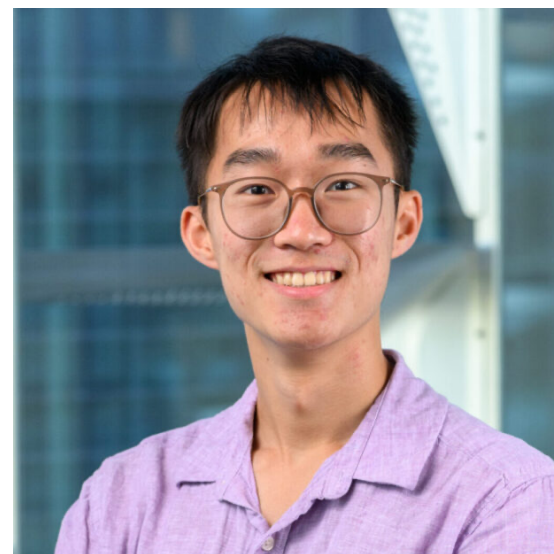
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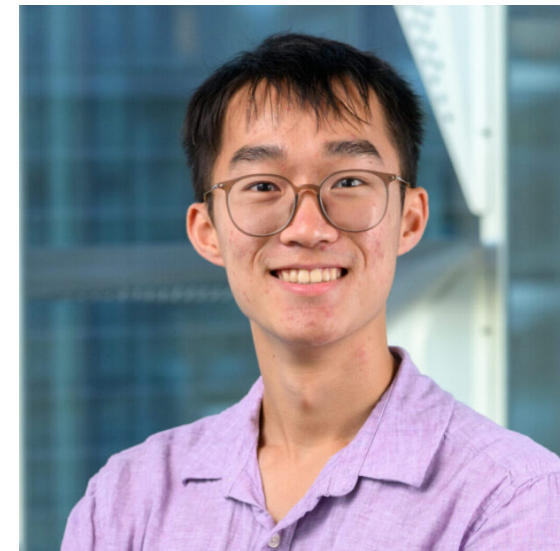
Egil Rolstad



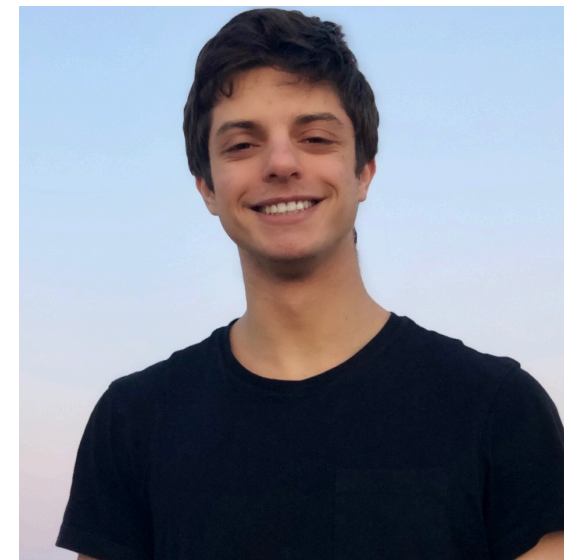
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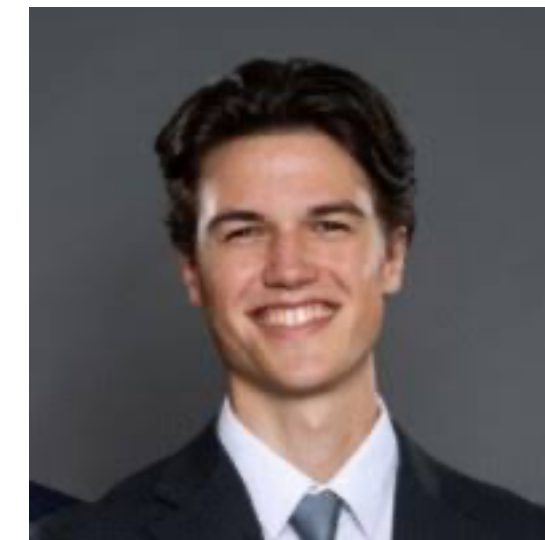
TFs: Alex Cai,



Philip Nielsen,



Egil Rolstad



CAs: Emmanuel Rassou,



Justin Li,



Ishaan Sinha



Announcements

Office Hours

website: <https://harvard-cs-1840.github.io/2026/>

Type	Day	Start	End	Staff	Campus	Location
Office Hours	Monday	3:45 PM	5:30 PM	Kianté	Allston	SEC 5.402
Office Hours	Monday	8:00 PM	9:30 PM	Justin	Cambridge	Mather Pset Nights
Office Hours	Thursday	2:15 PM	4:00 PM	Emmanuel	Allston	SEC 2.122/2.123
Office Hours	Thursday	4:30 PM	6:15 PM	Alex	Cambridge	Cruft Lab 309
Office Hours	Tuesday	3:45 PM	5:30 PM	Egil	Allston	SEC 2.122/2.123
Office Hours	Tuesday	7:00 PM	8:45 PM	Philip	Cambridge	Pierce Hall 301
Office Hours	Wednesday	7:00 PM	8:45 PM	Ishaan	Cambridge	Winthrop CS Nights
Section	Monday	2:15 PM	3:30 PM	Alex	Allston	SEC 1.412
Section	Thursday	12:45 PM	2:00 PM	Emmanuel	Allston	SEC 1.412
Section	Tuesday	11:15 AM	12:30 PM	Egil	Allston	SEC 1.412
Section	Tuesday	4:30 PM	5:45 PM	Philip	Cambridge	Pierce Hall 213 (Brooks Room)
Section	Wednesday	4:30 PM	5:45 PM	Ishaan & Justin	Cambridge	Maxwell Dworkin G135

Announcements

Homework 1 released 9/11 **(this Friday)**

Homework 0 due 9/17 **(next Friday)**

If any part of Homework 0 feels challenging or unfamiliar,
please reach out to the TFs/CAs for help.

Reading Materials

Three Books:

1) Reinforcement Learning: Theory and Algorithms

by Alekh Agarwal, Kianté Brantley, Nan Jiang, Sham Kakade, and Wen Sun.

This is an advanced RL theory book and is mostly out of scope for this class.

Please let us know if you find any typos or mistakes in the book

2) An Introduction to Reinforcement Learning

by Alex Cai

3) Reinforcement Learning: An Introduction

by Richard Sutton and Andrew Barto.

This is the classic text in reinforcement learning.

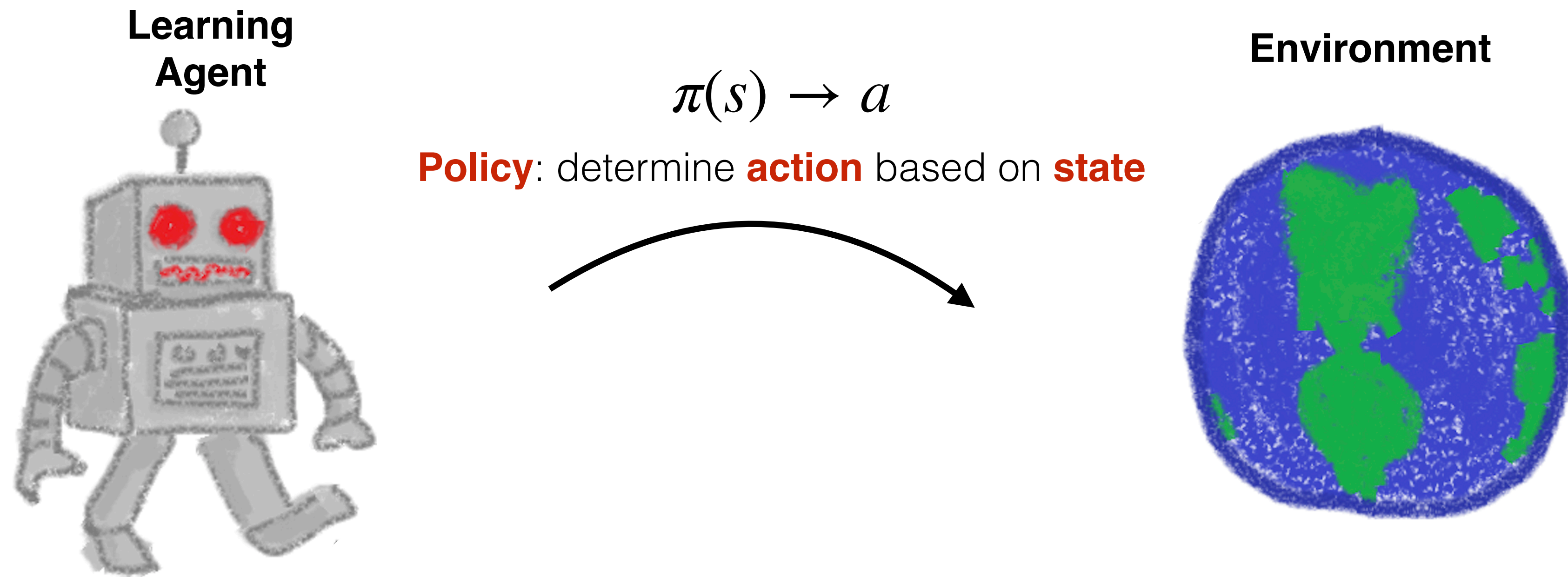
Questions?

Outline:

1. Definitions of Markov Decision Process
2. Value functions (V and Q functions)
3. Bellman equations

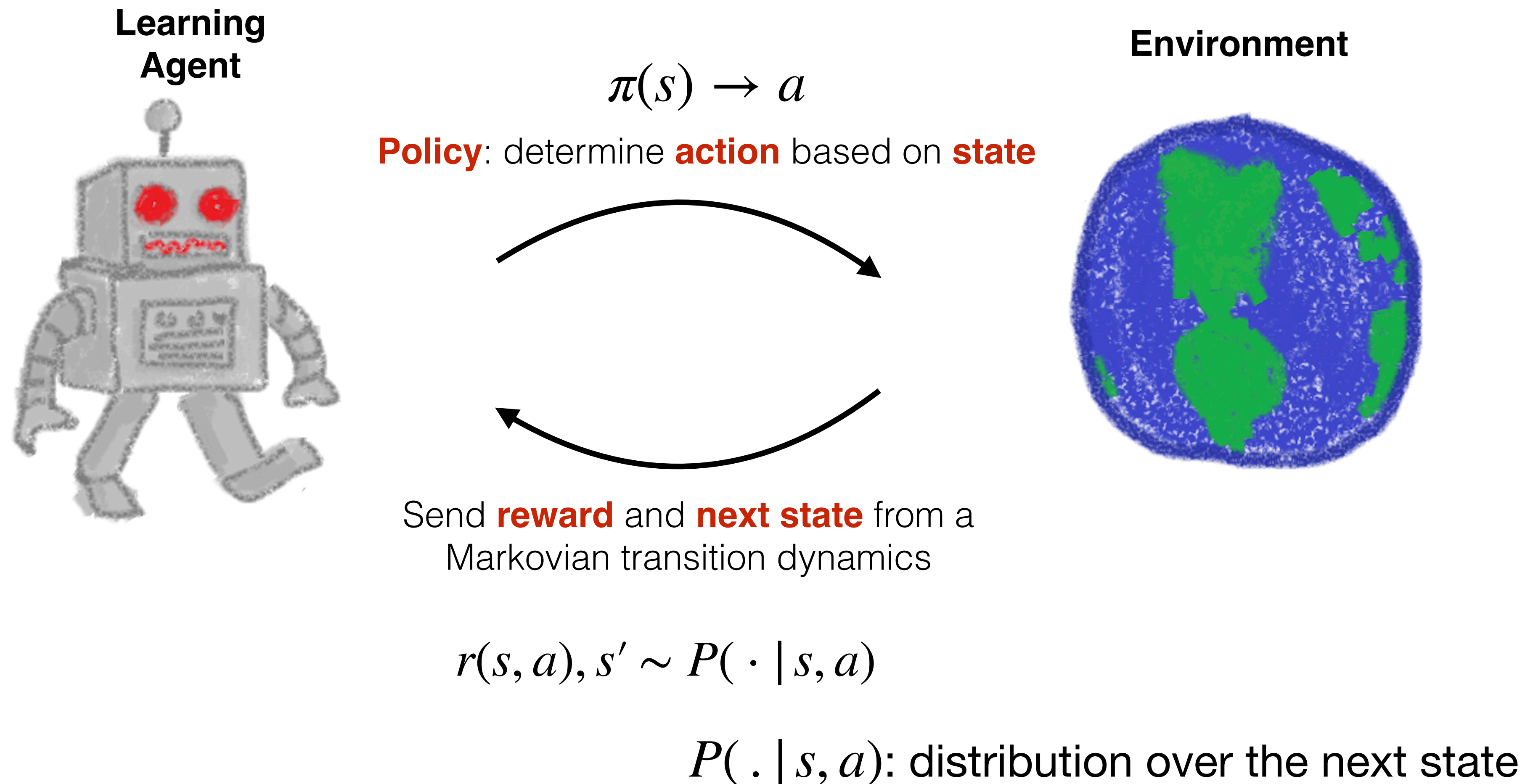
The Mathematical framework:

Infinite horizon Markov Decision Process



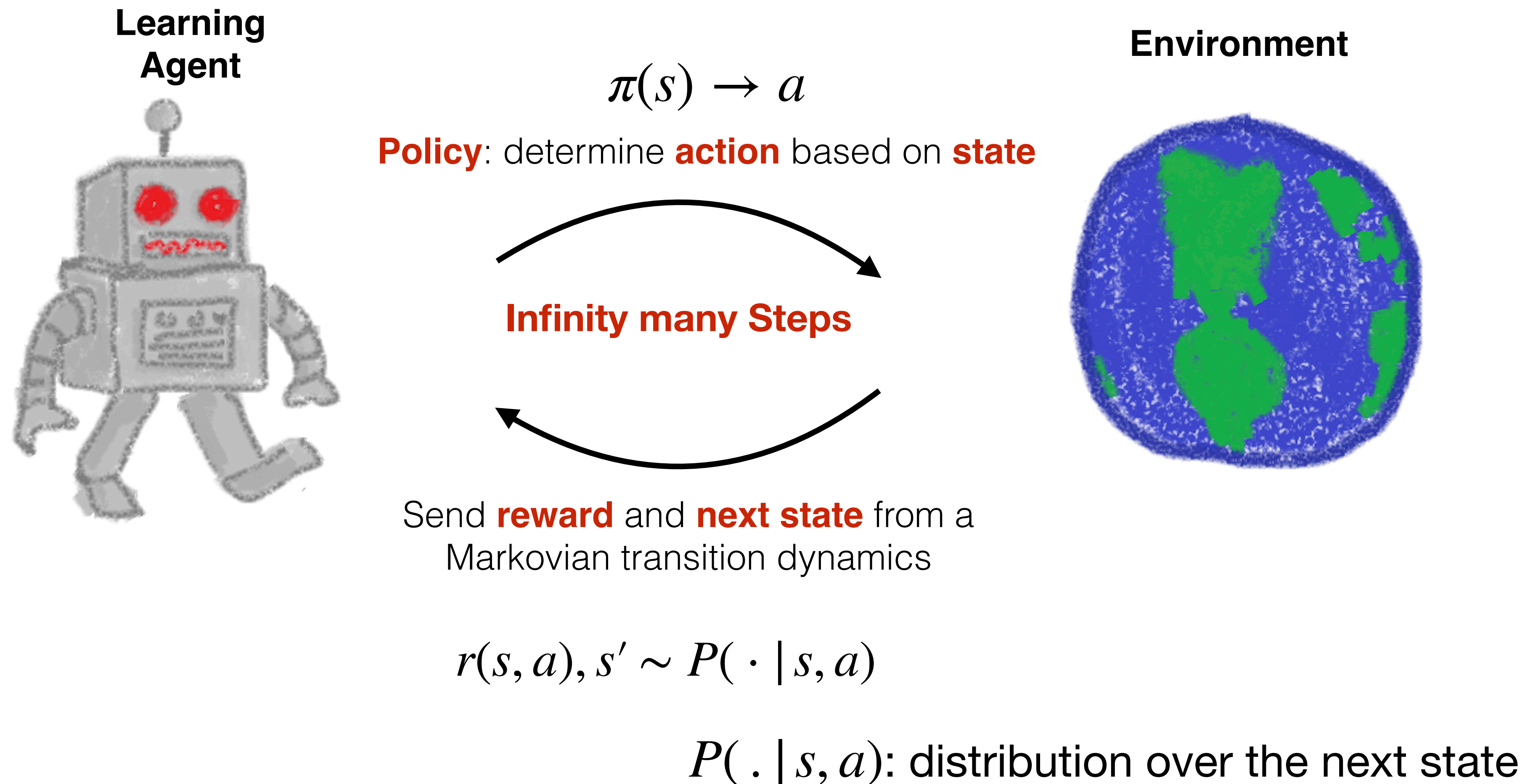
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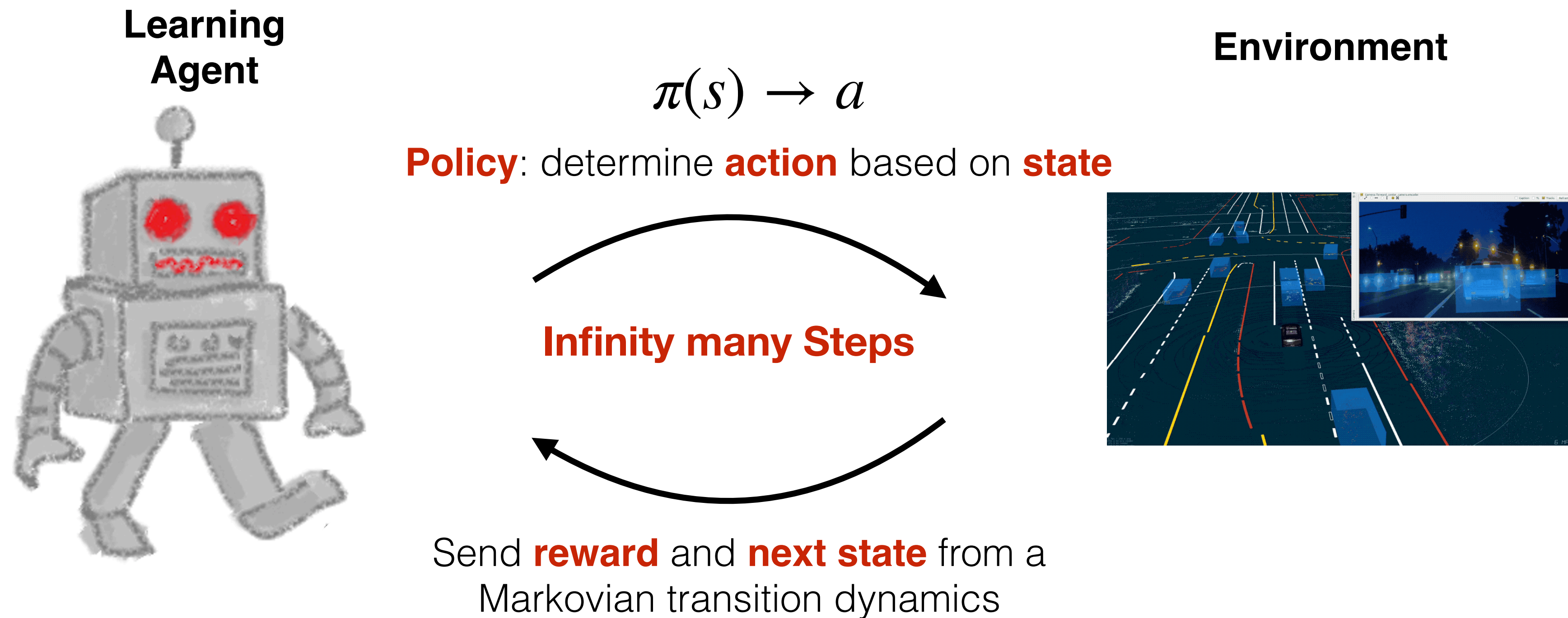
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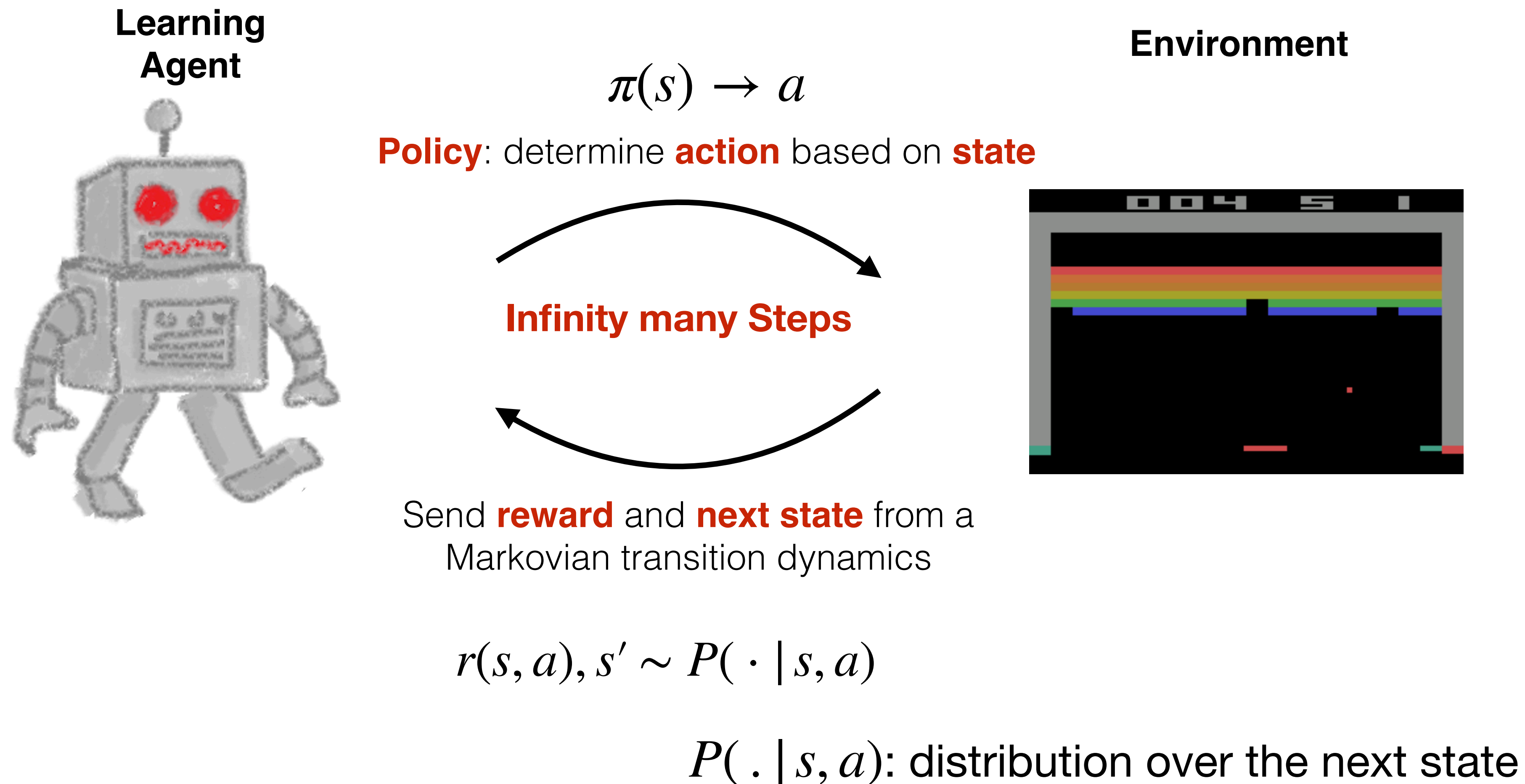


$$r(s, a), s' \sim P(\cdot | s, a)$$

$P(\cdot | s, a)$: distribution over the next state

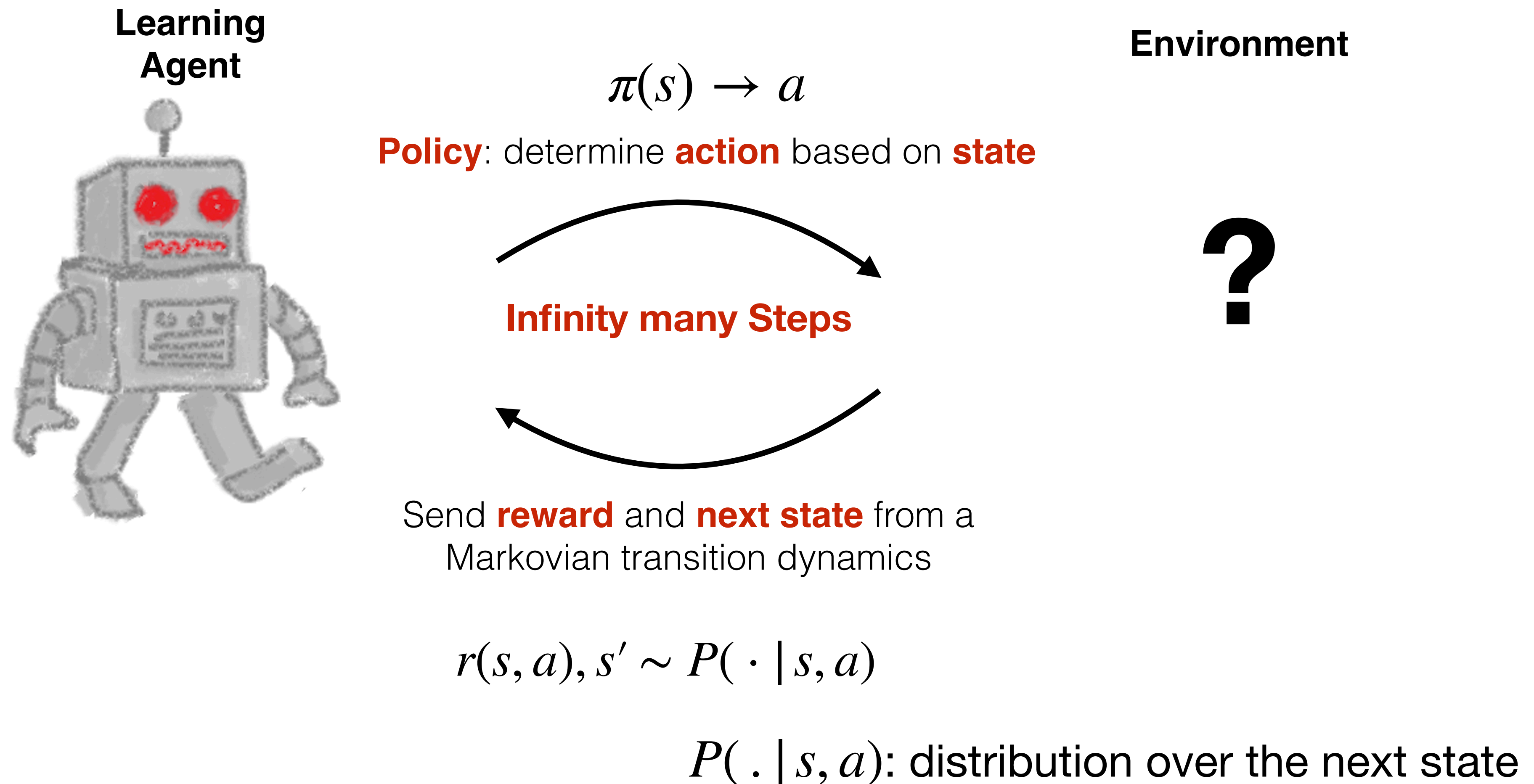
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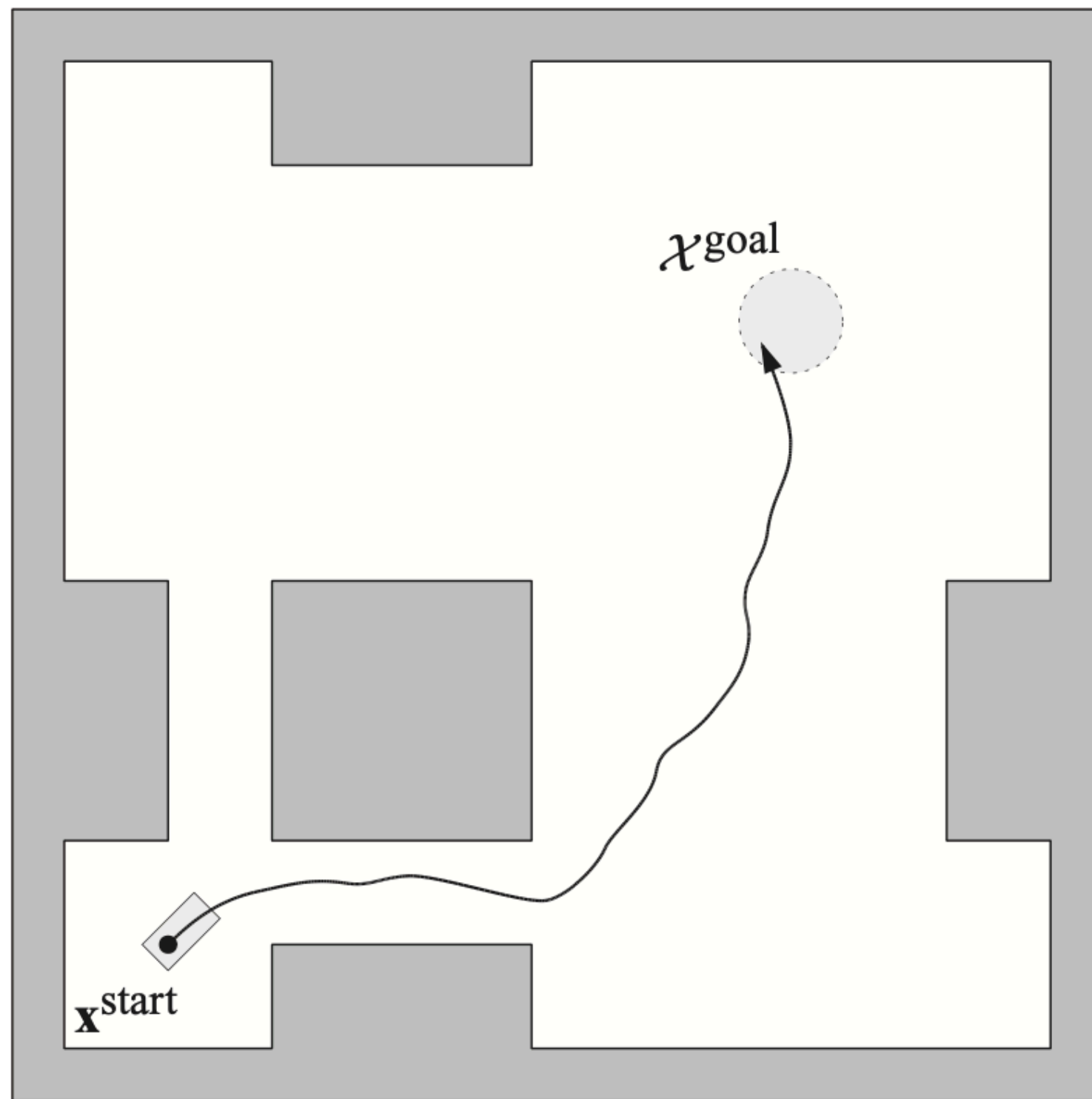


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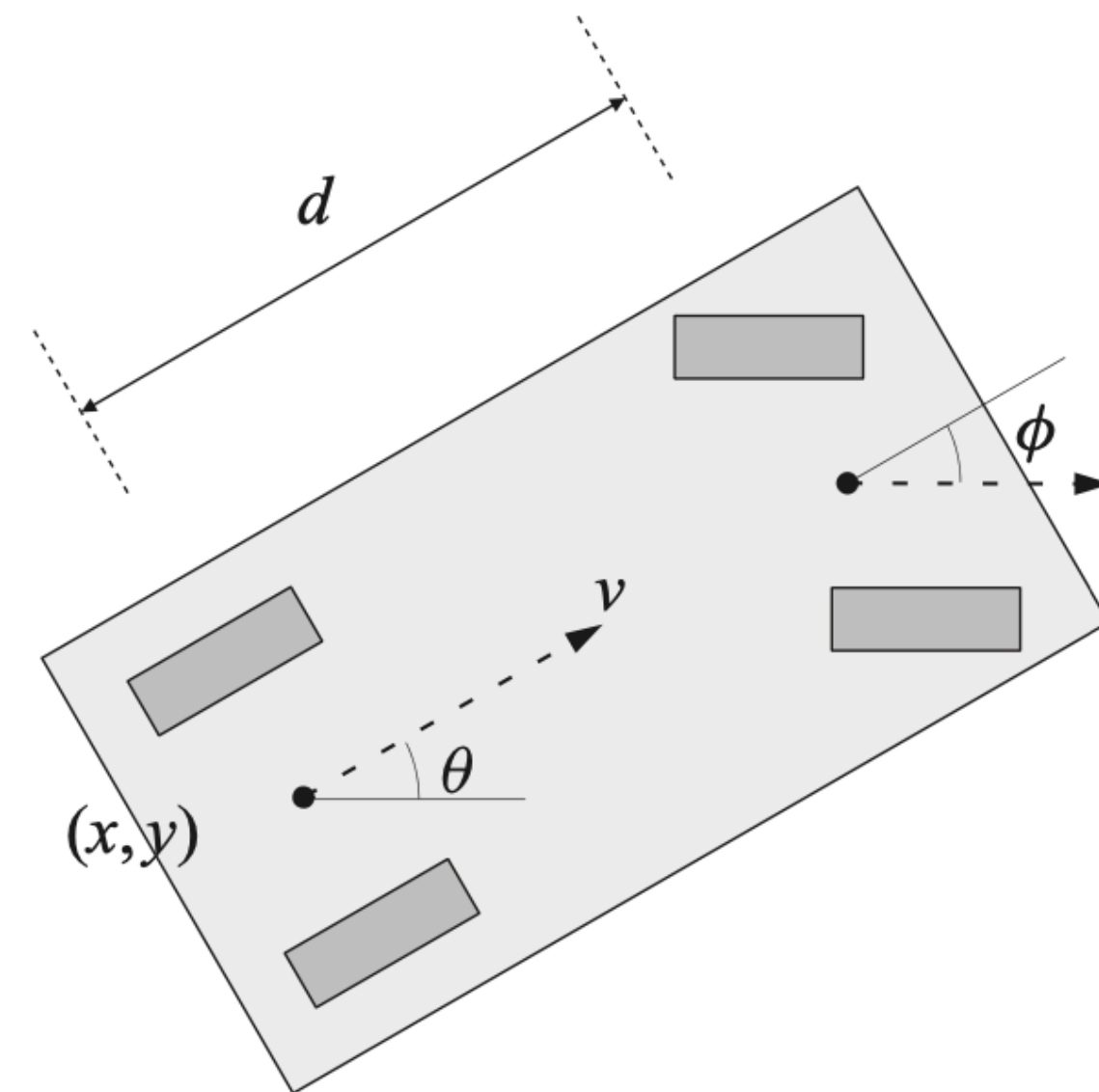
Infinite horizon Markov Decision Process



Example: 2-D simple car navigation



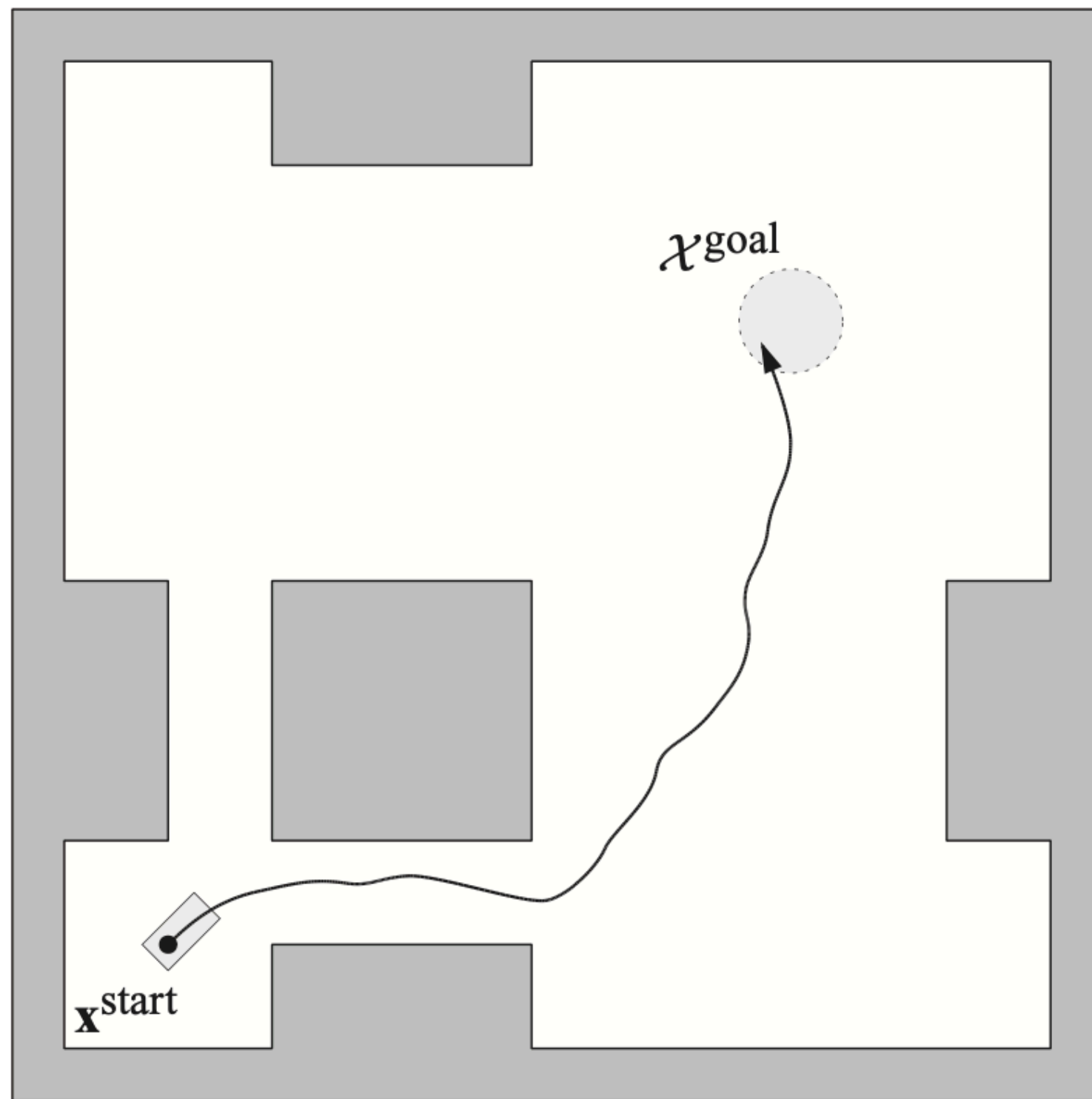
(a)



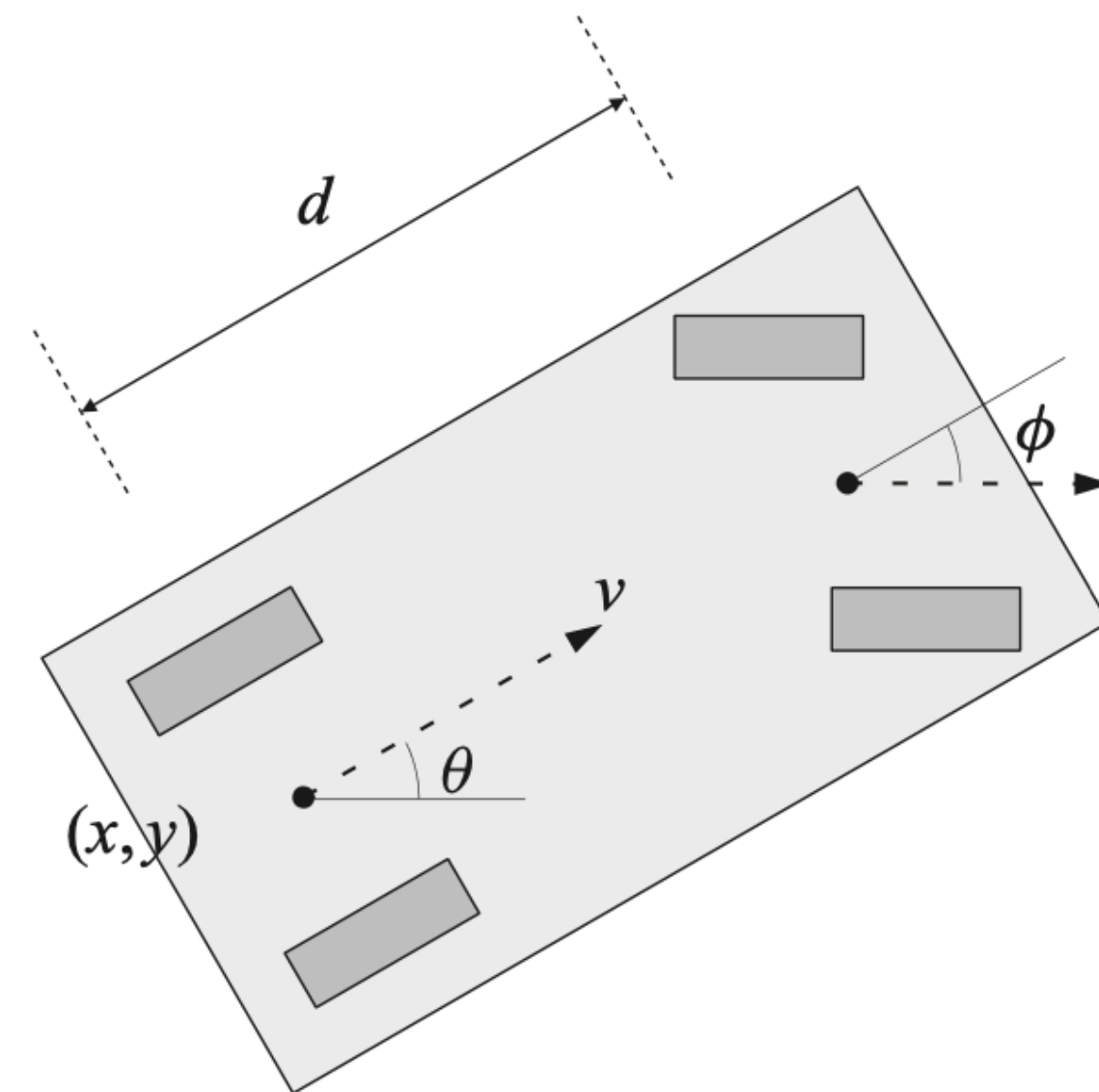
(b)

Example: 2-D simple car navigation

$$s = [x, y, \theta, v]^T \in \mathbb{R}^4$$



(a)

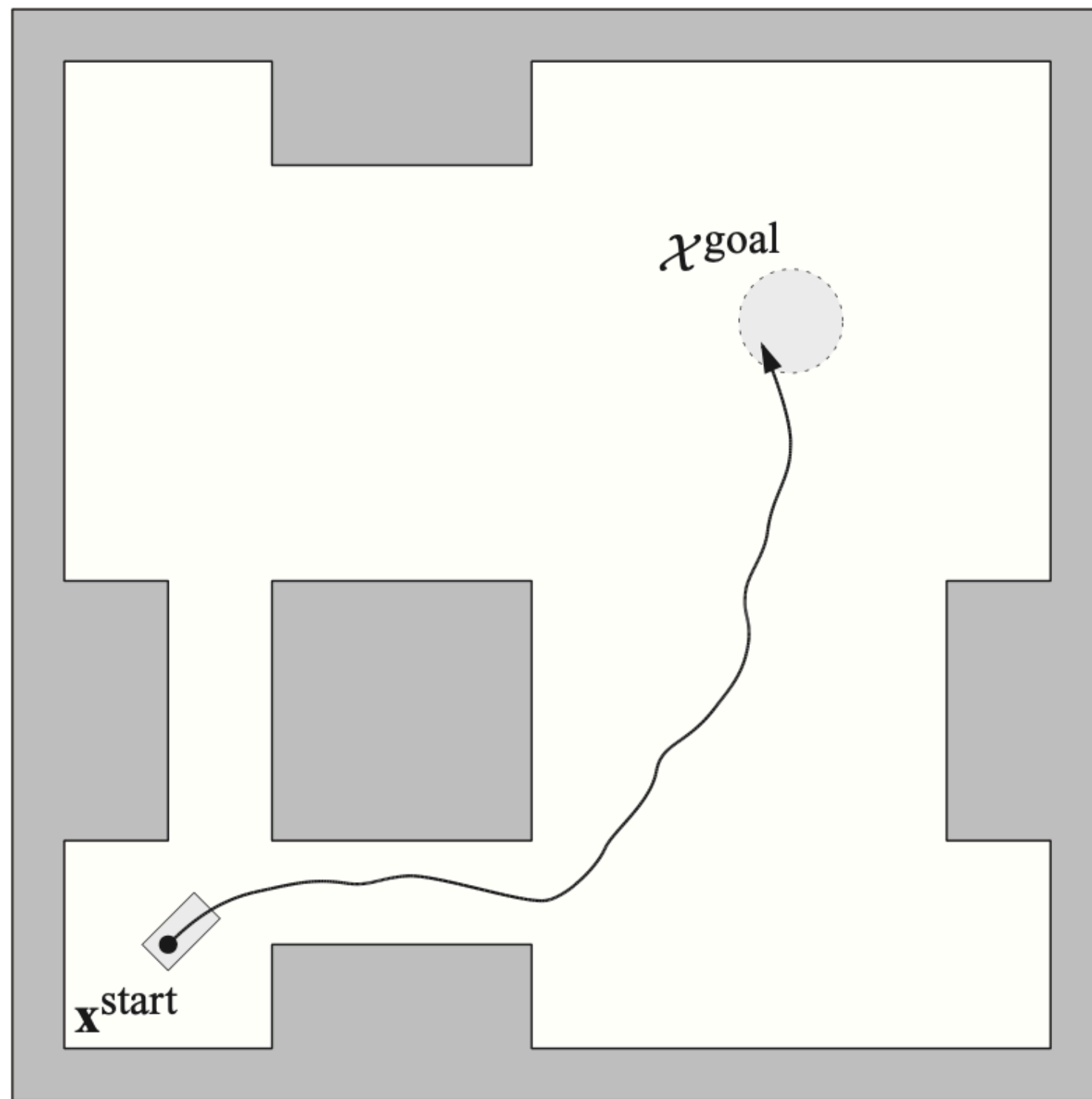


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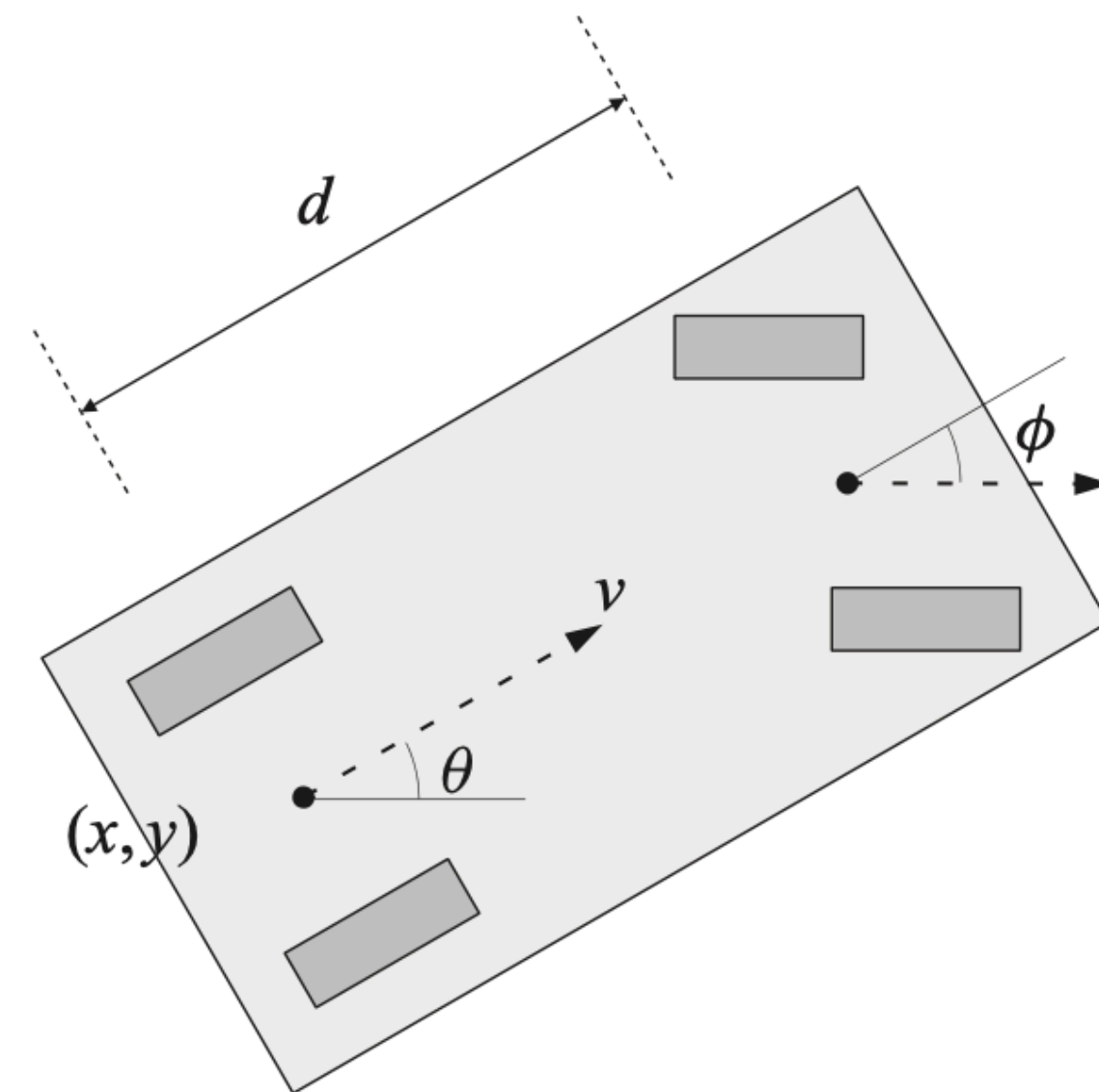
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$$a = [\alpha, \phi]^T \in \mathbb{R}^2$$

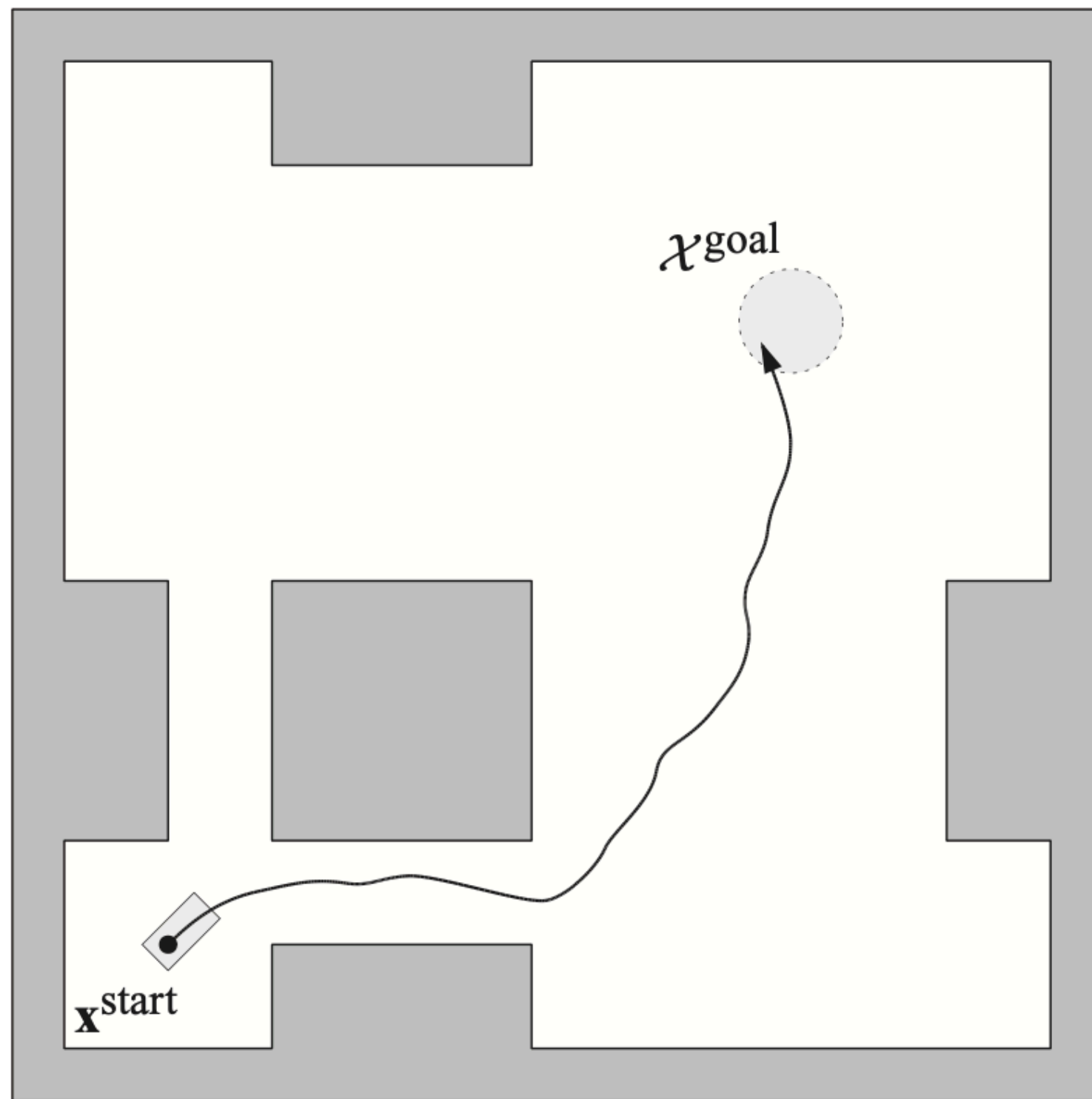


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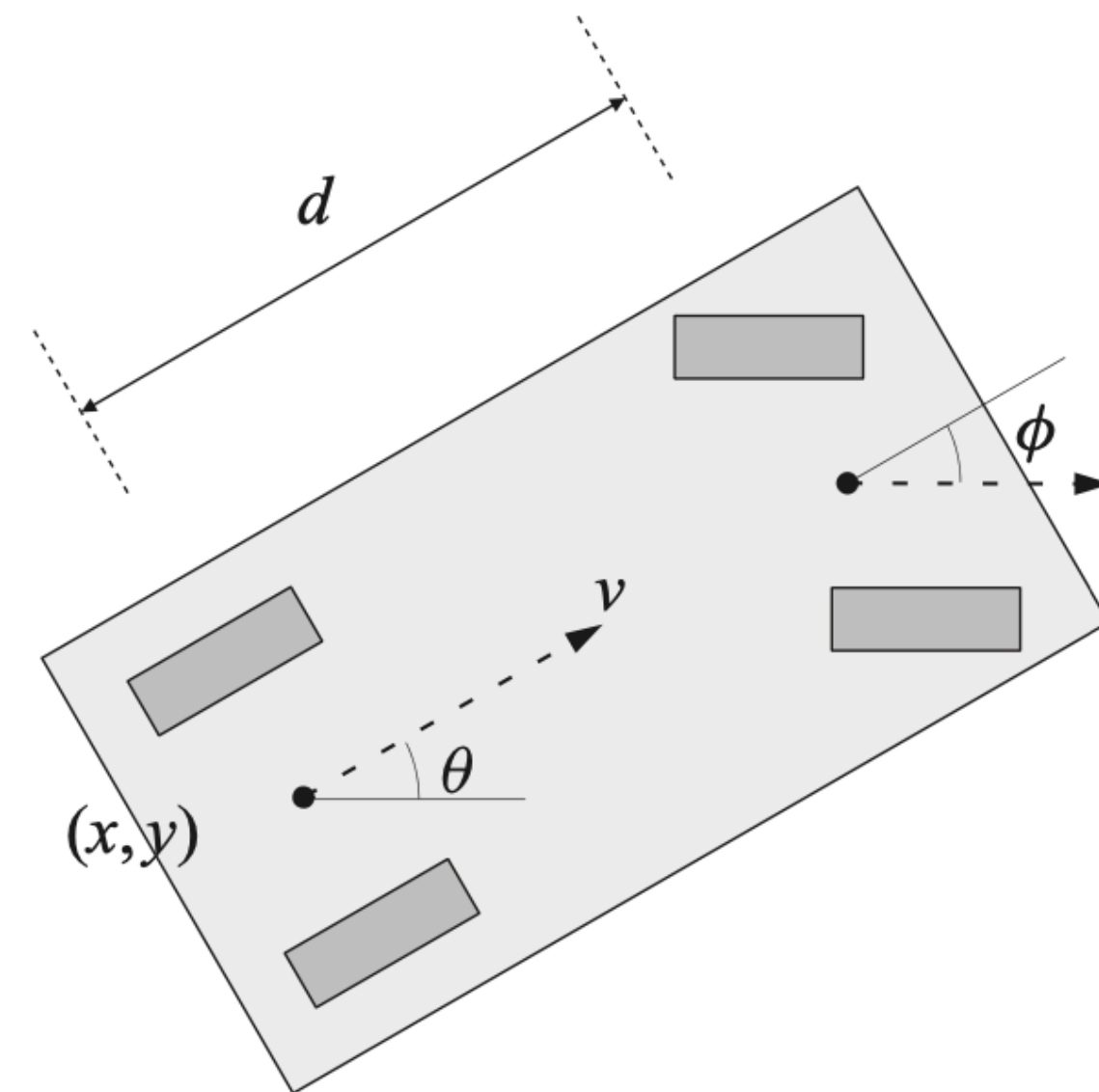


(b)

Example: 2-D simple car navigation



(a)



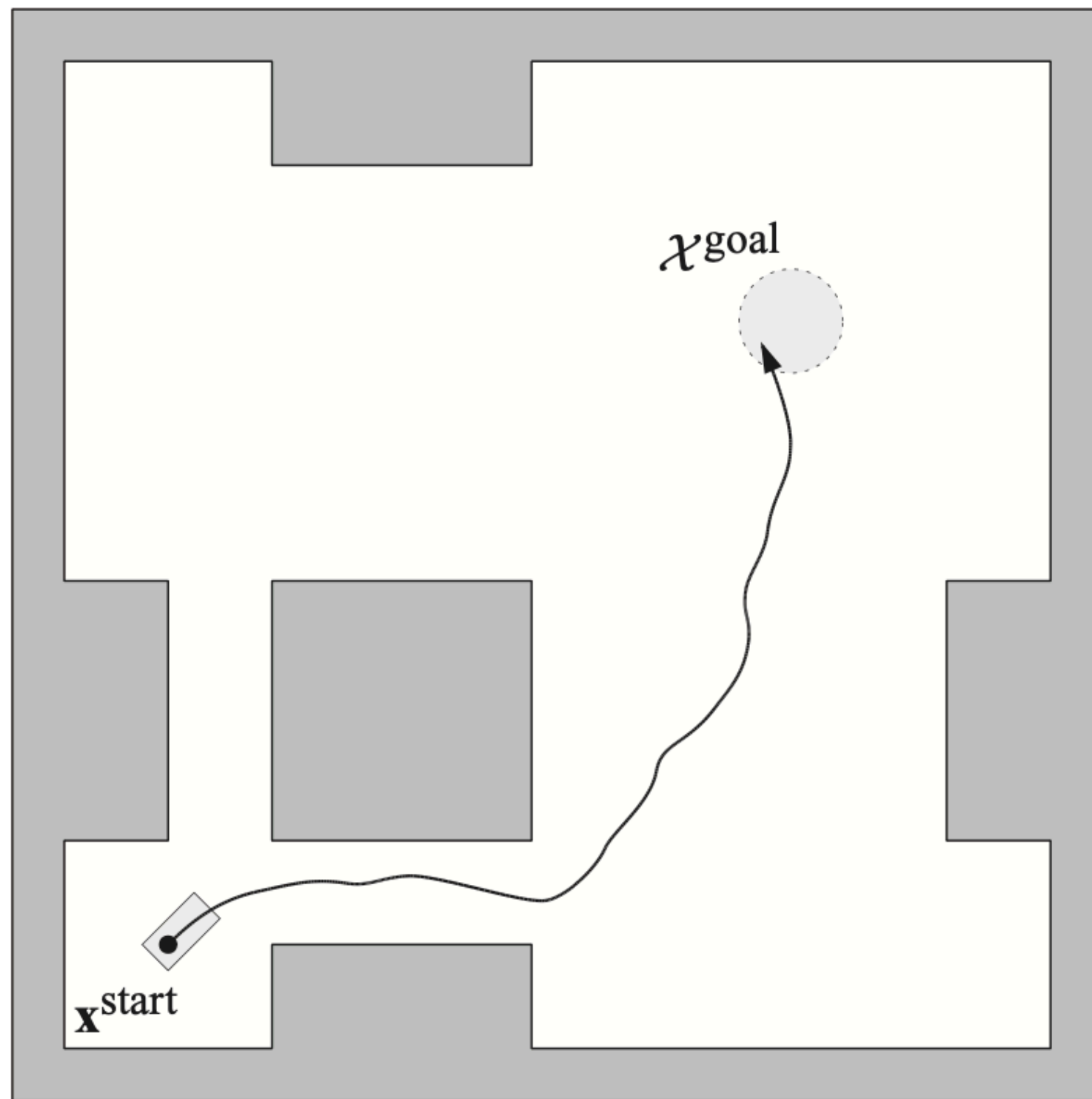
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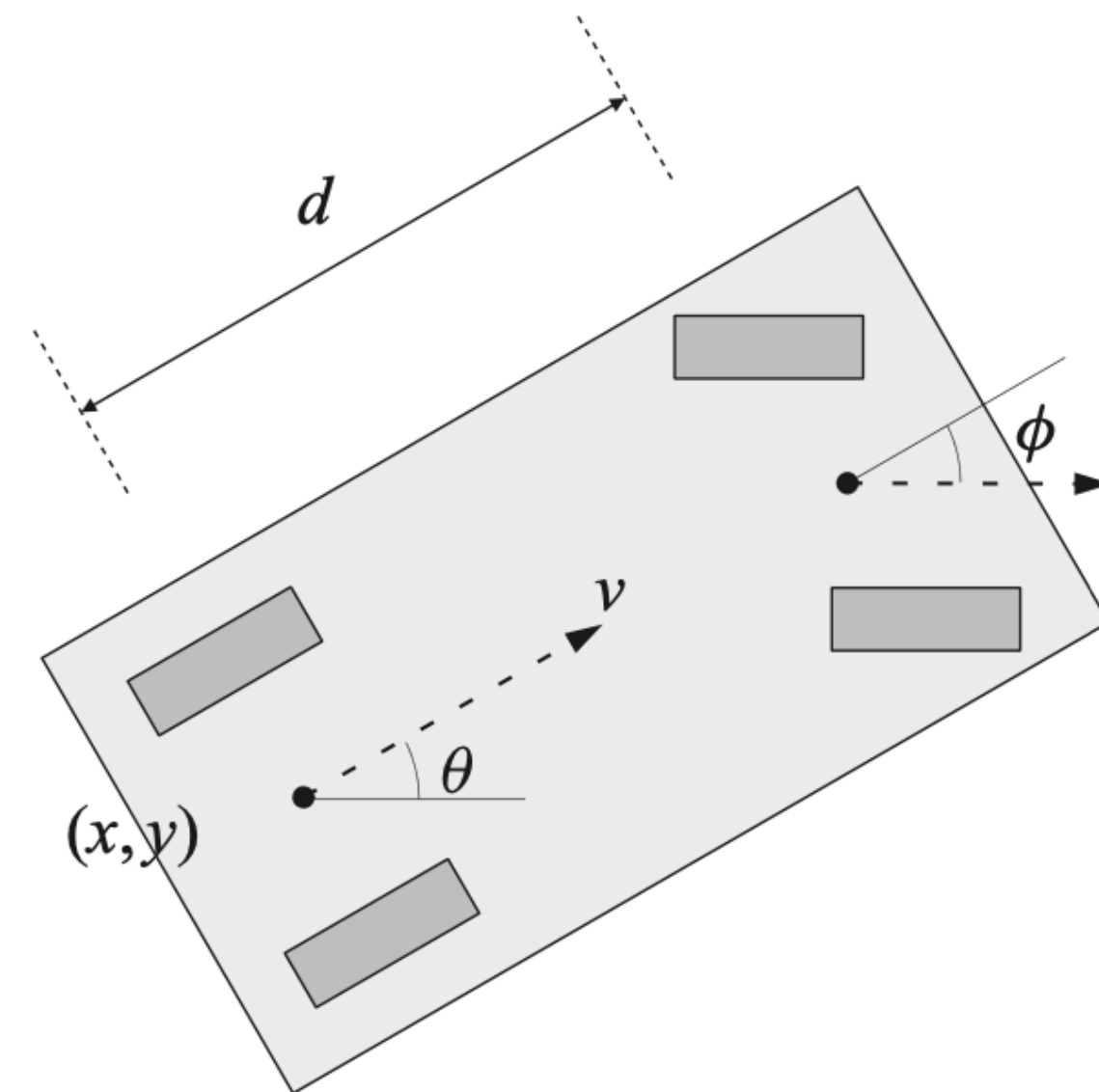
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$$r(s, a) = \begin{cases} 100 & (x, y) \in \mathcal{X}_{\text{goal}} \\ -1 & \text{hit obstacles} \\ 0 & \text{else} \end{cases}$$

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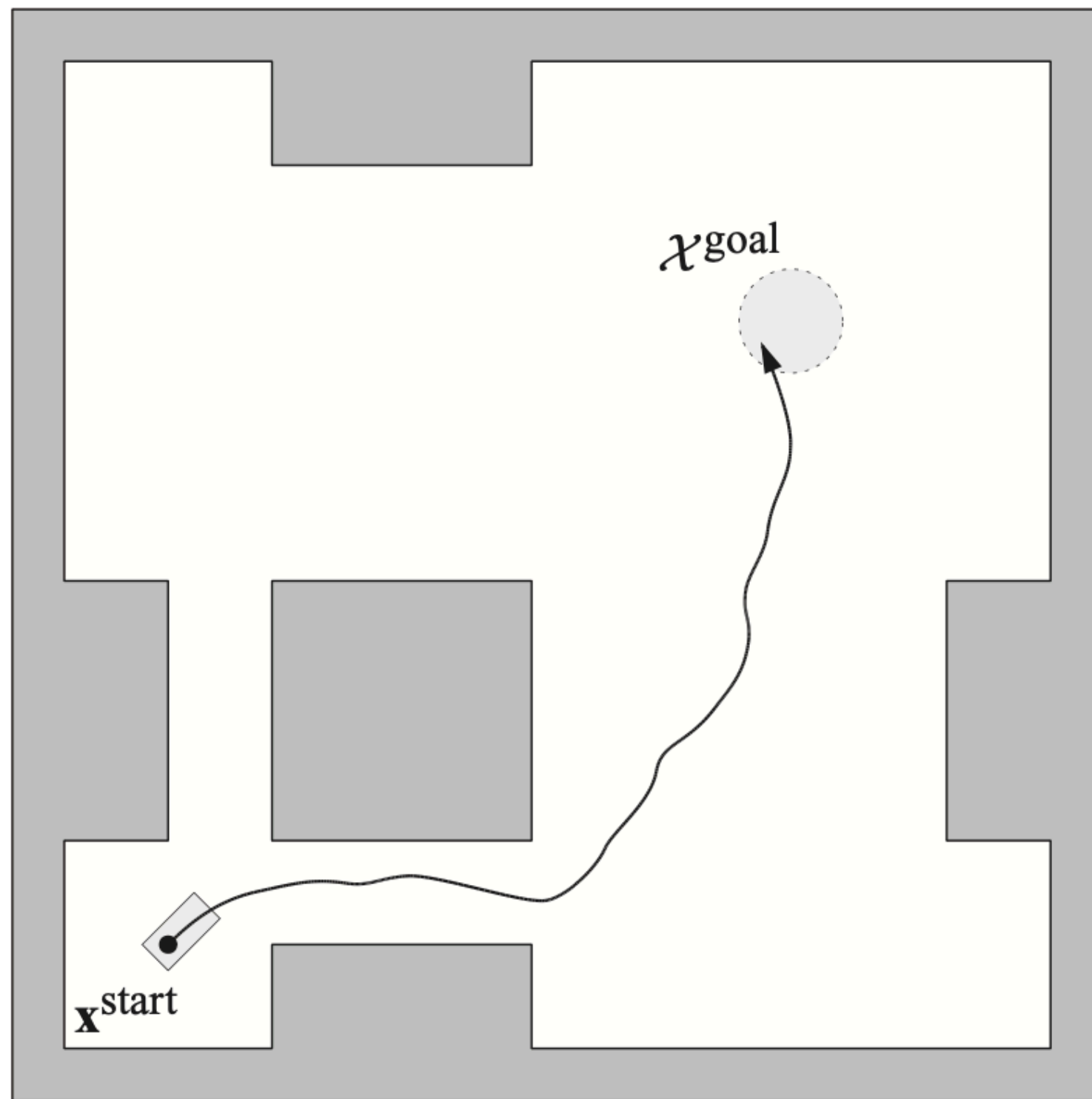
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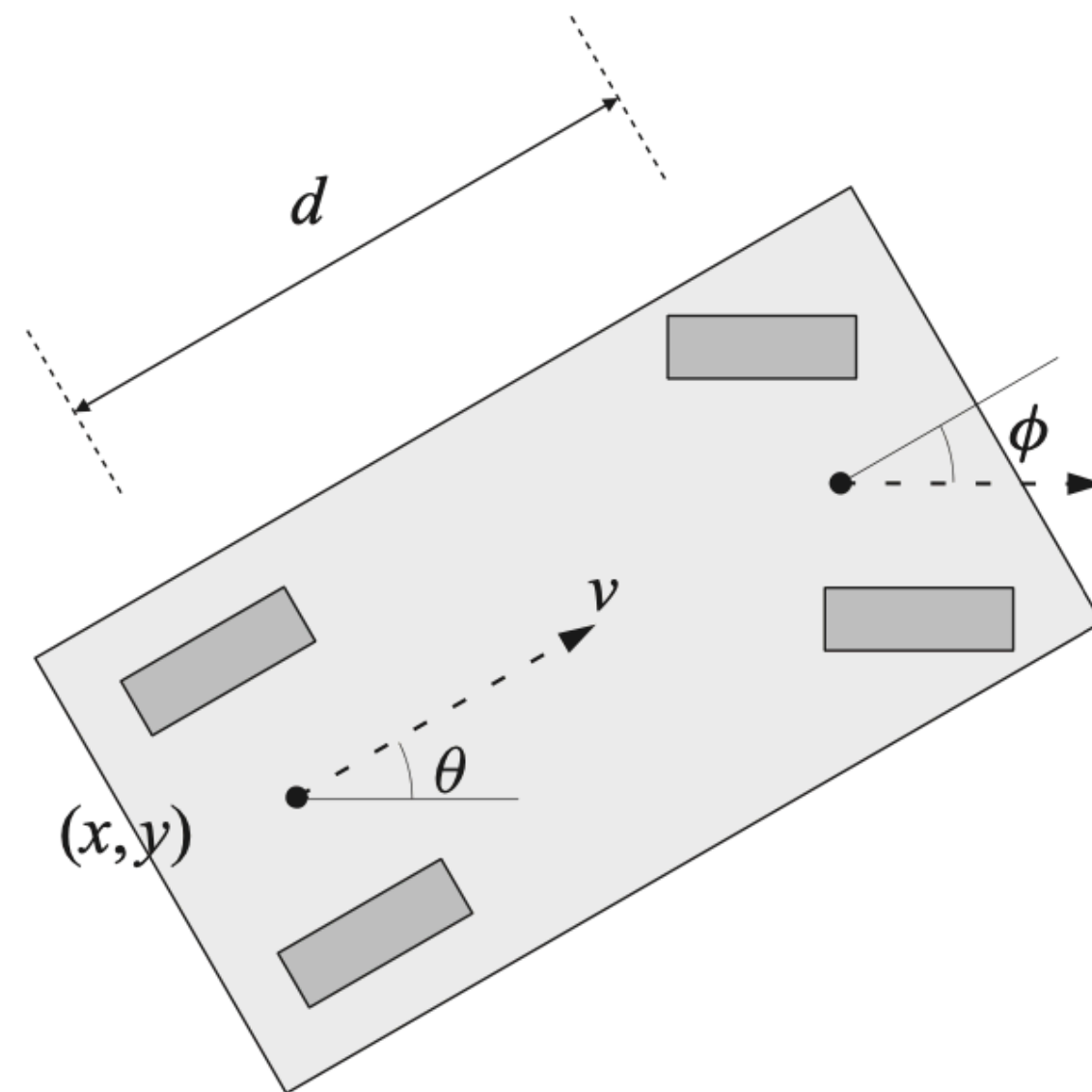
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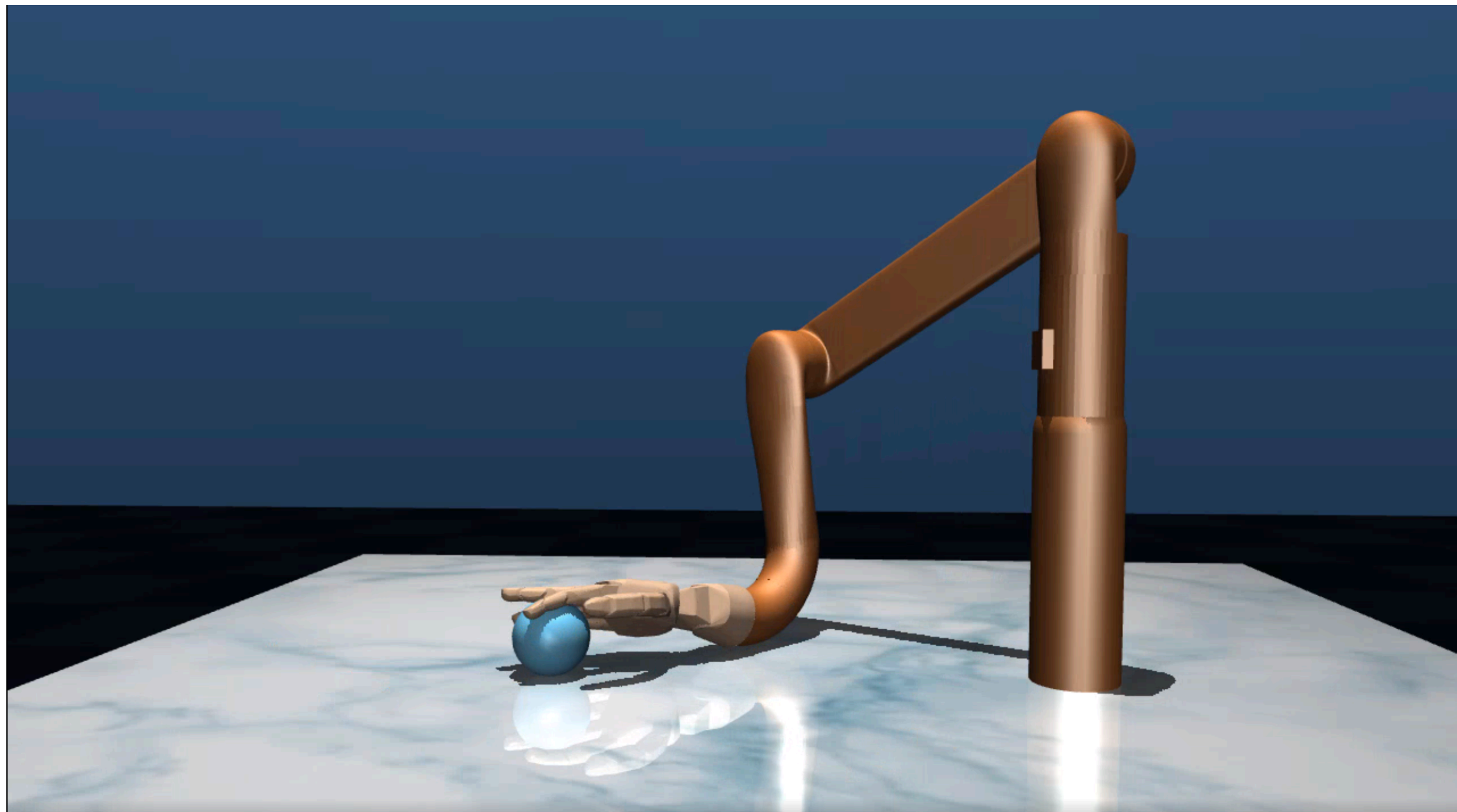
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$$f(s, a) = \begin{bmatrix} x + \tau v \cos \theta \\ y + \tau v \sin \theta \\ \theta + \tau v \tan(\phi)/d \\ v + \tau \alpha \end{bmatrix}$$

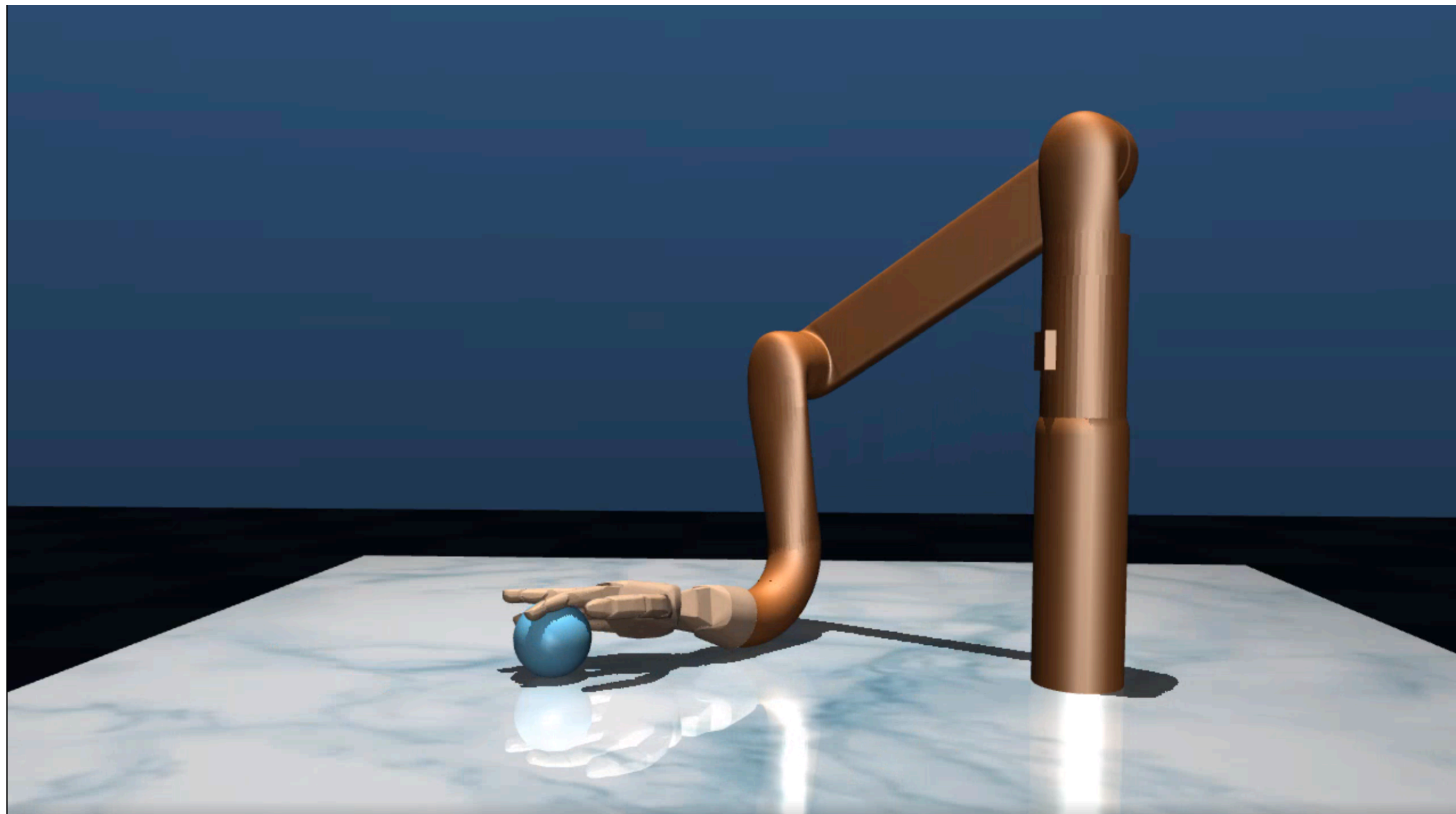
Example: Robotic Pick-and-Place

robot hand needs to pick the ball and hold it in a goal (x,y,z) position



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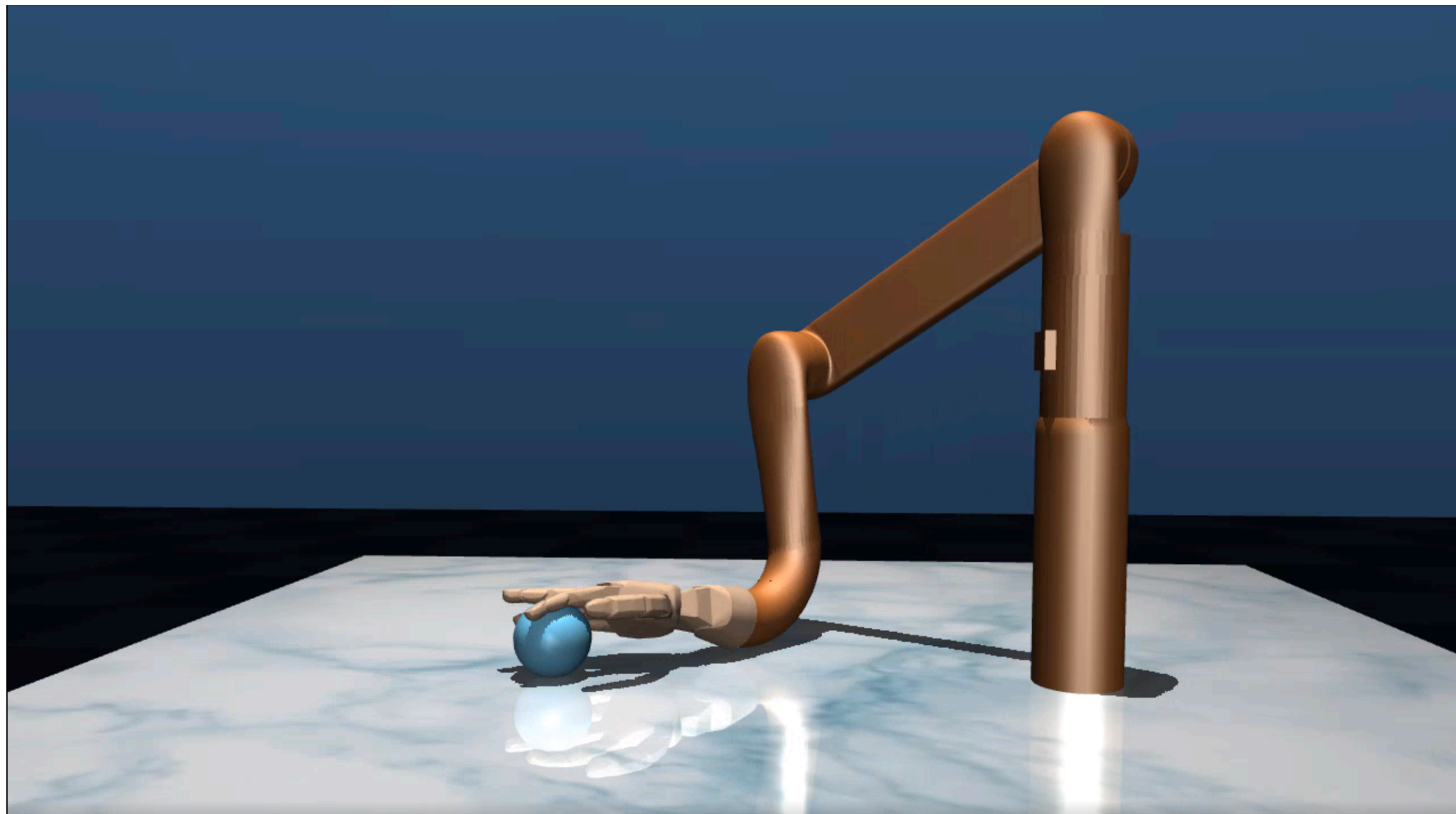
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and the ball's position

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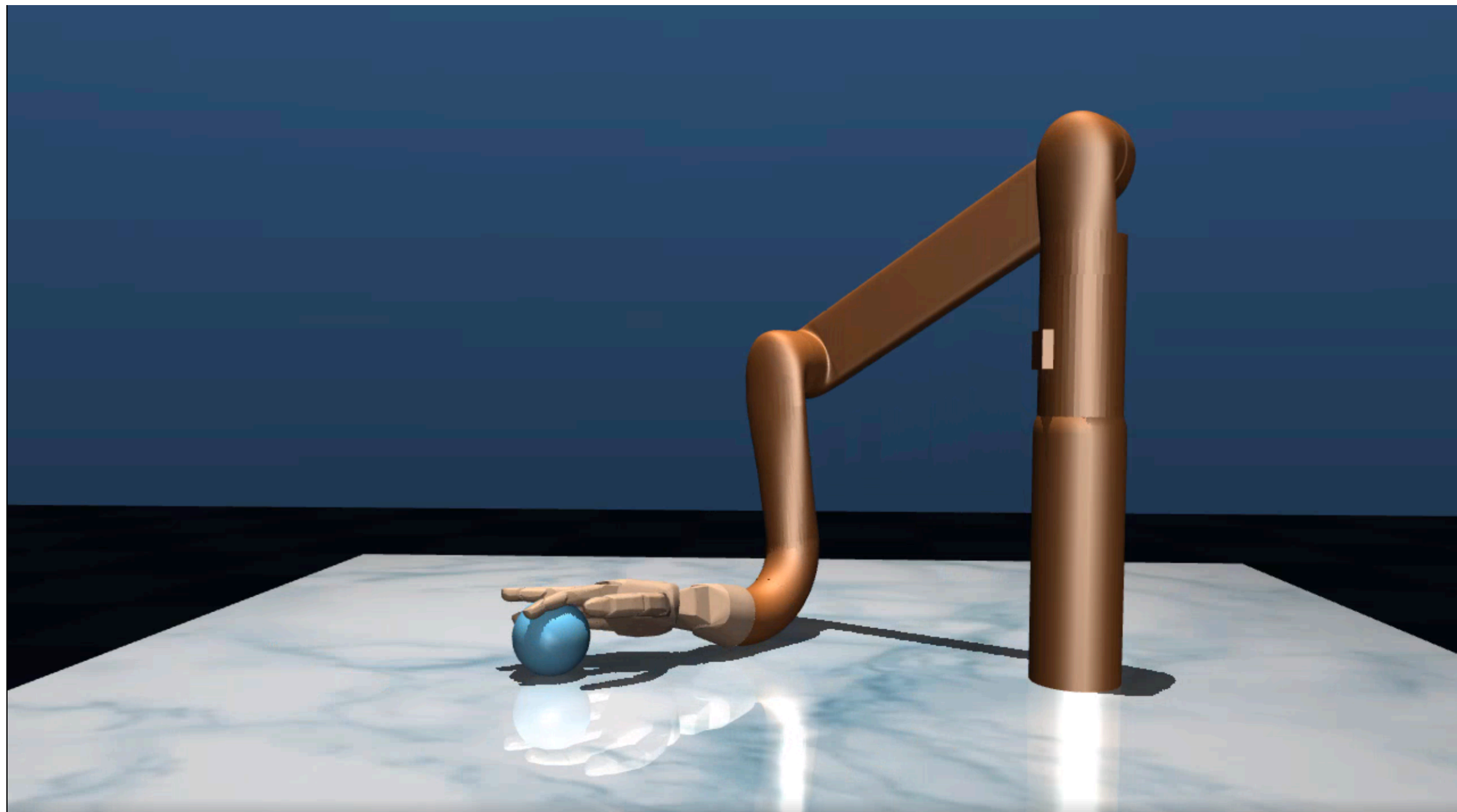


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Action a : Torque on joints in arm & fingers

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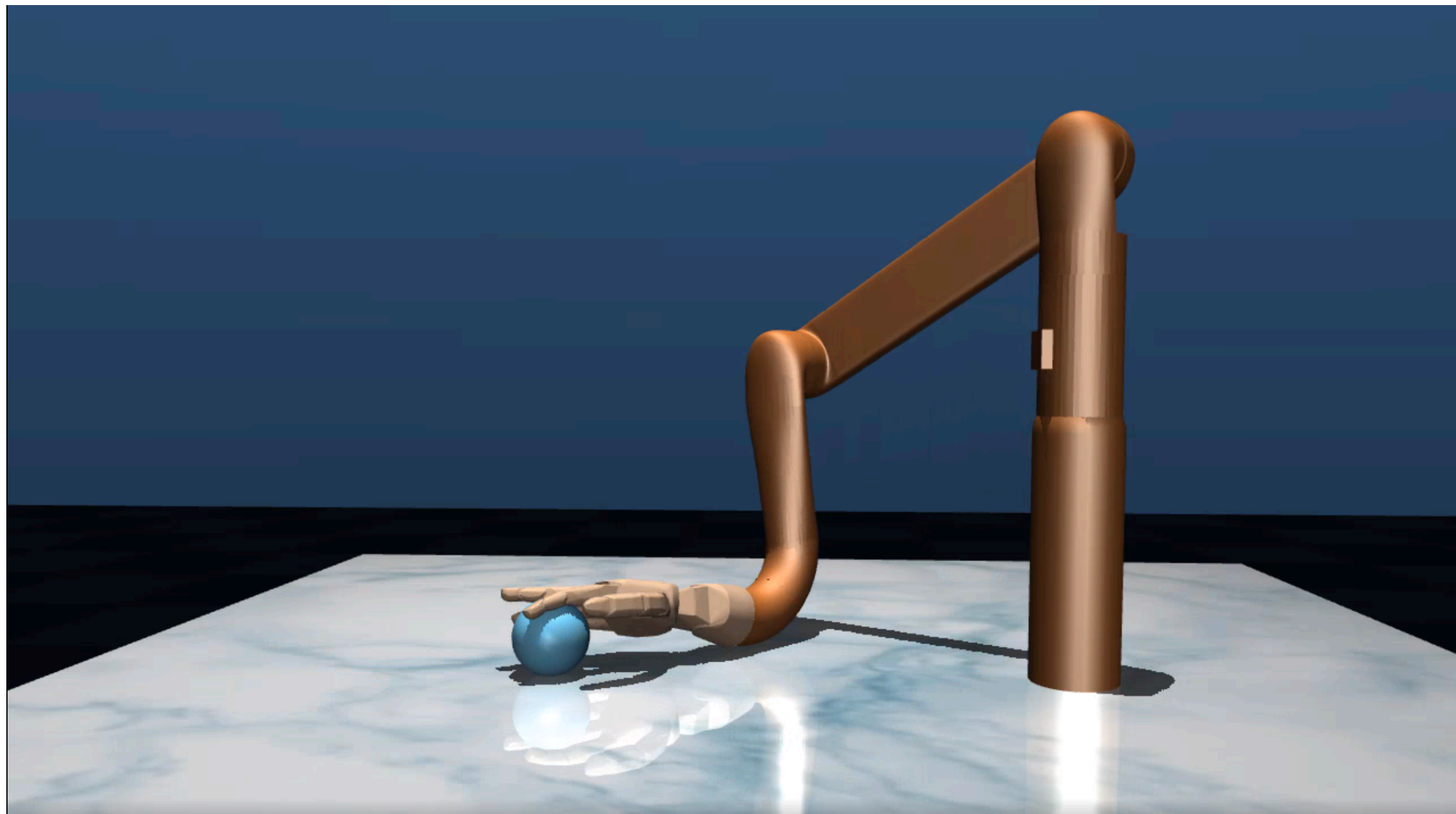
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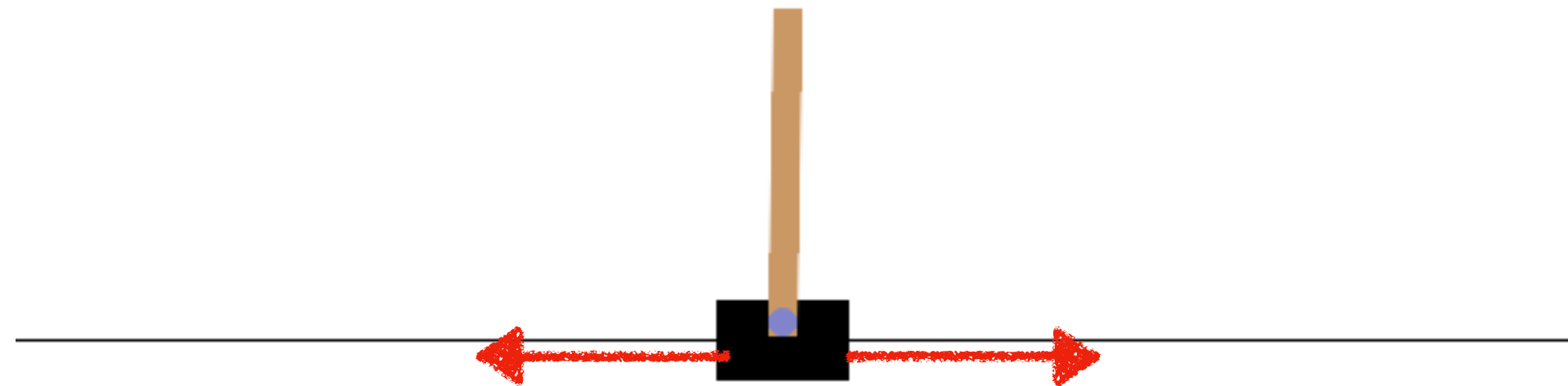
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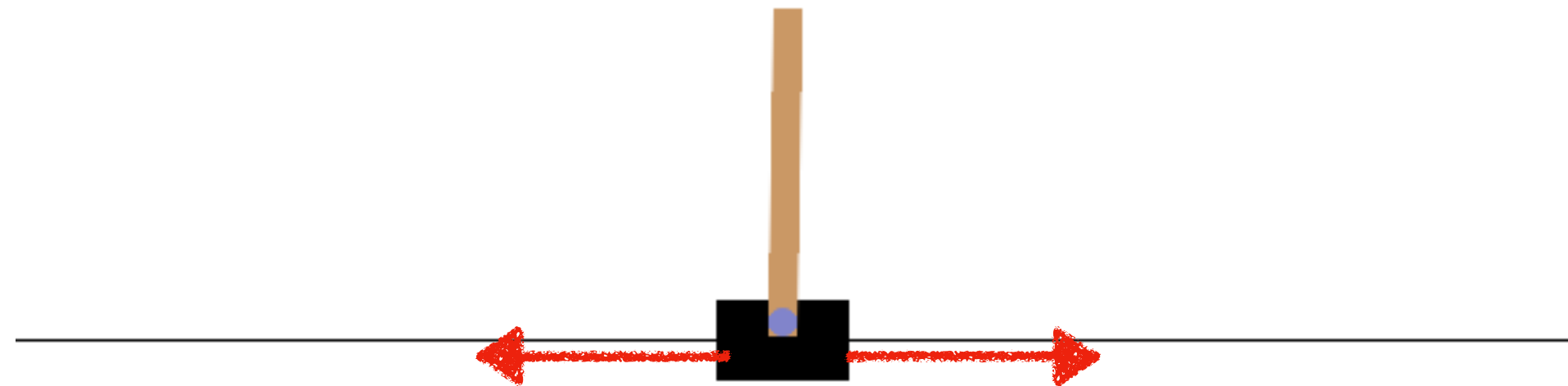
Transition $s' \sim P(\cdot | s, a)$: physics + some noise

Cost $c(s, a)$: torque magnitude + dist to goal

Example: OpenAI Gym demonstrations

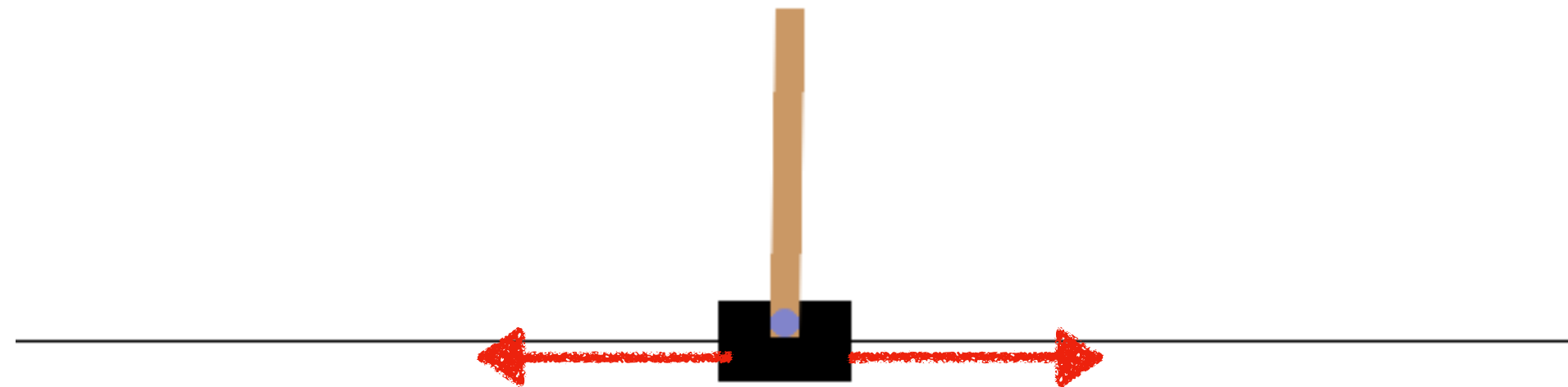


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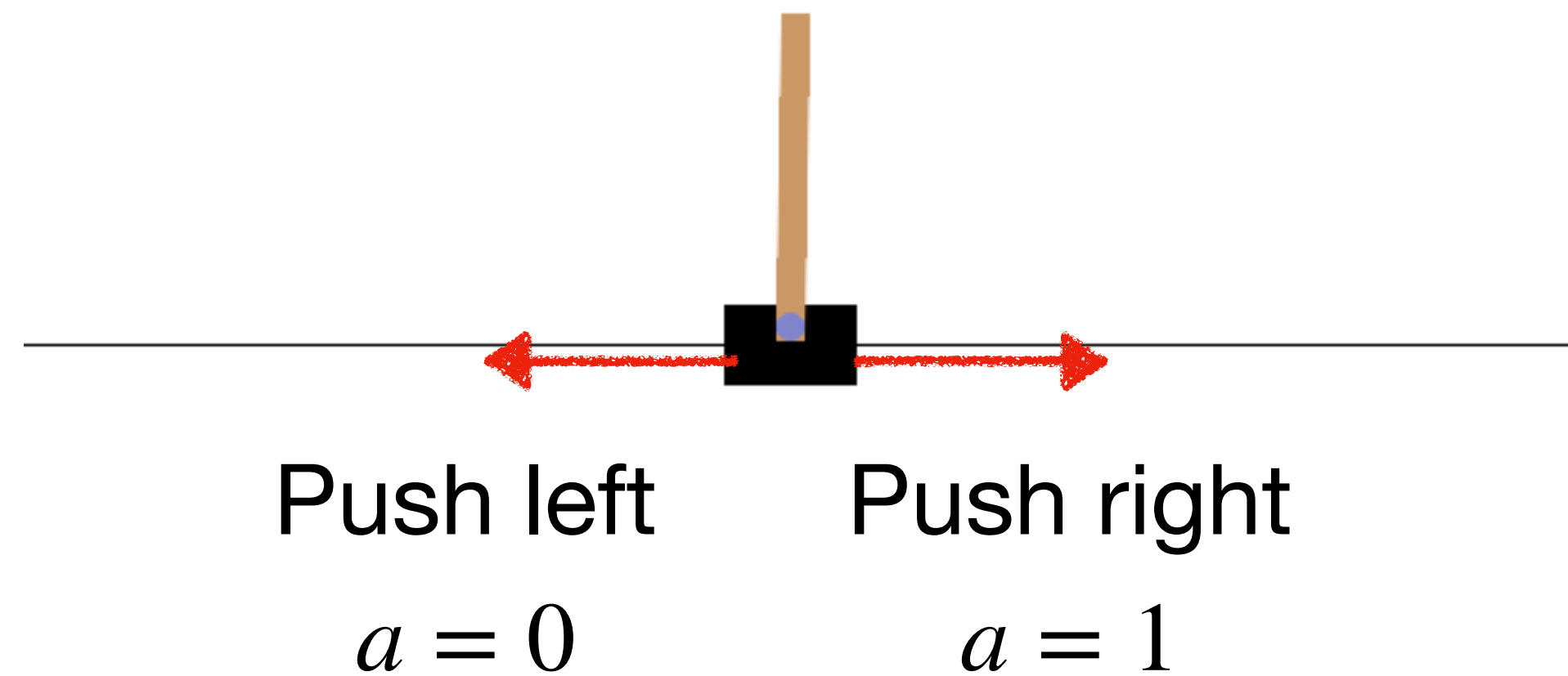
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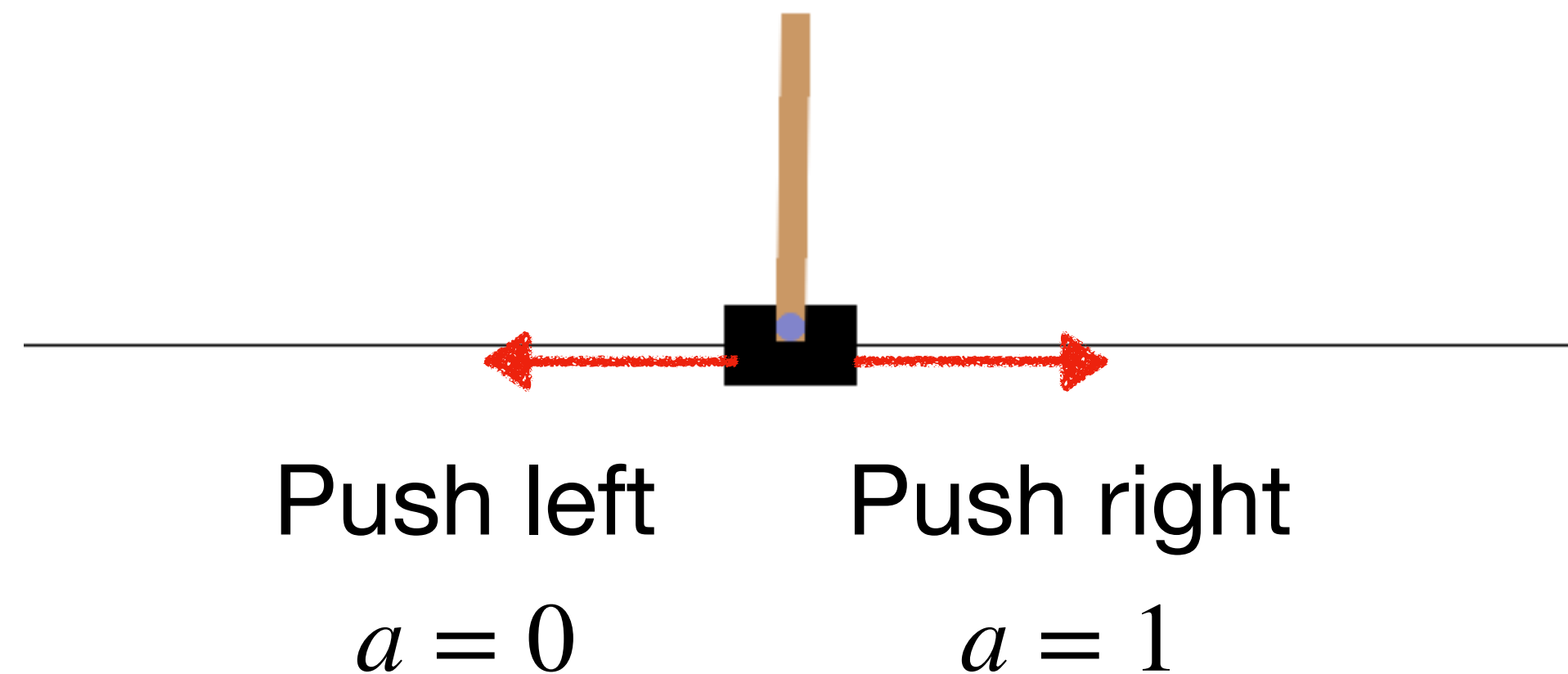
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$$r(s, a) = \begin{cases} 1 & \text{pole angle} \in [-12^\circ, 12^\circ], \\ 0 & \text{else} \end{cases}$$

Policy

$$\mathcal{S} \rightarrow \Delta(\mathcal{A})$$

A mapping from state to action (what action should I take if I'm in this state...)

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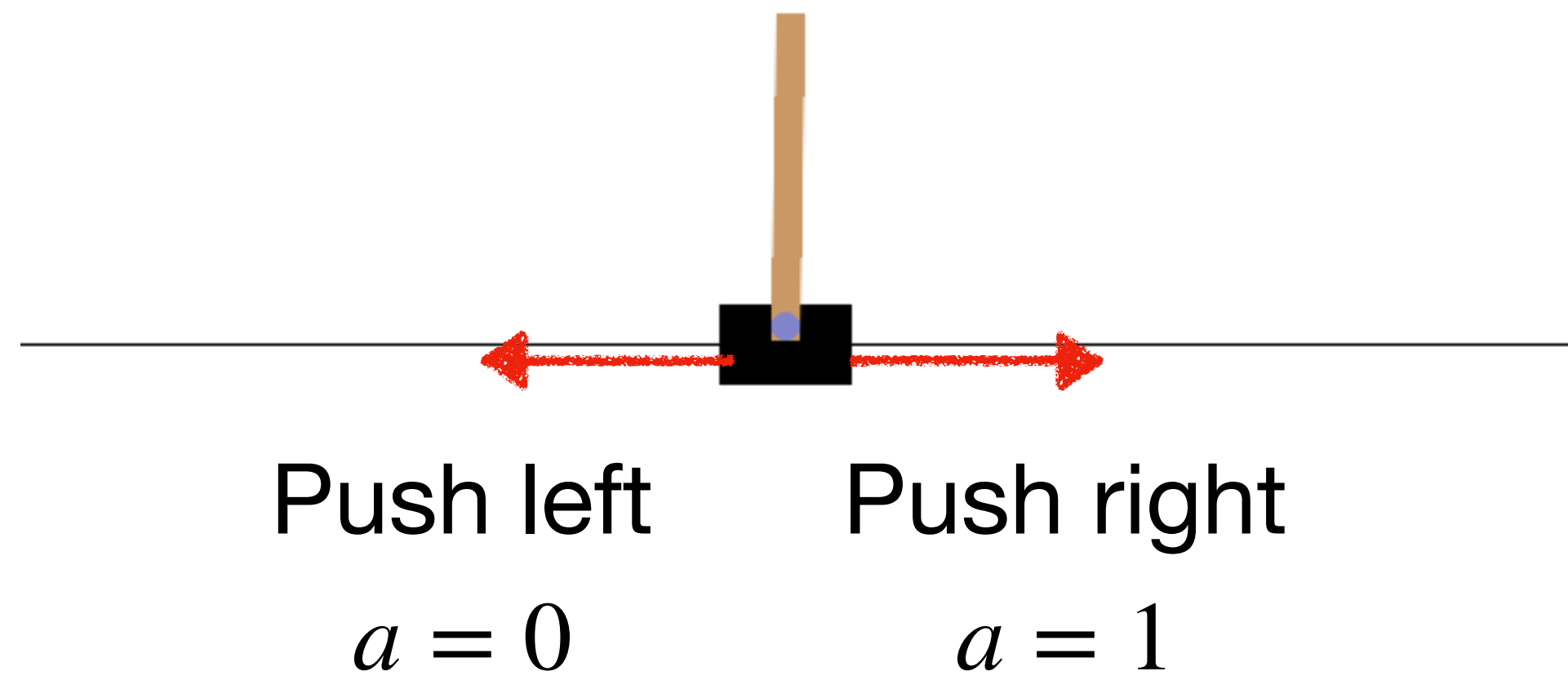
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Deterministic vs stochastic?

Q: Assume S state and A actions, how many different deterministic policies we can have?

Example: OpenAI Gym demonstrations

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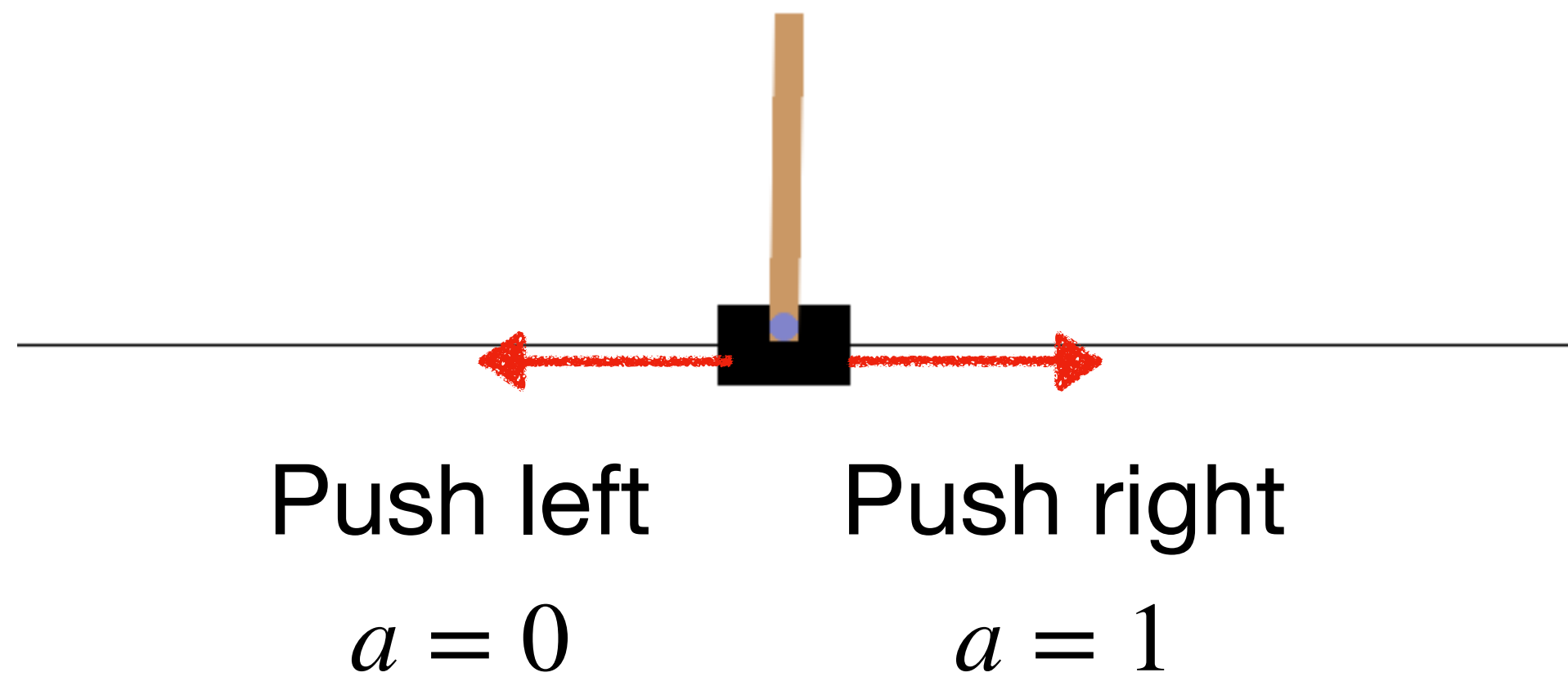


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Policy 1: uniform random
 $\pi(0 | s) = \pi(1 | s) = 0.5, \forall s$

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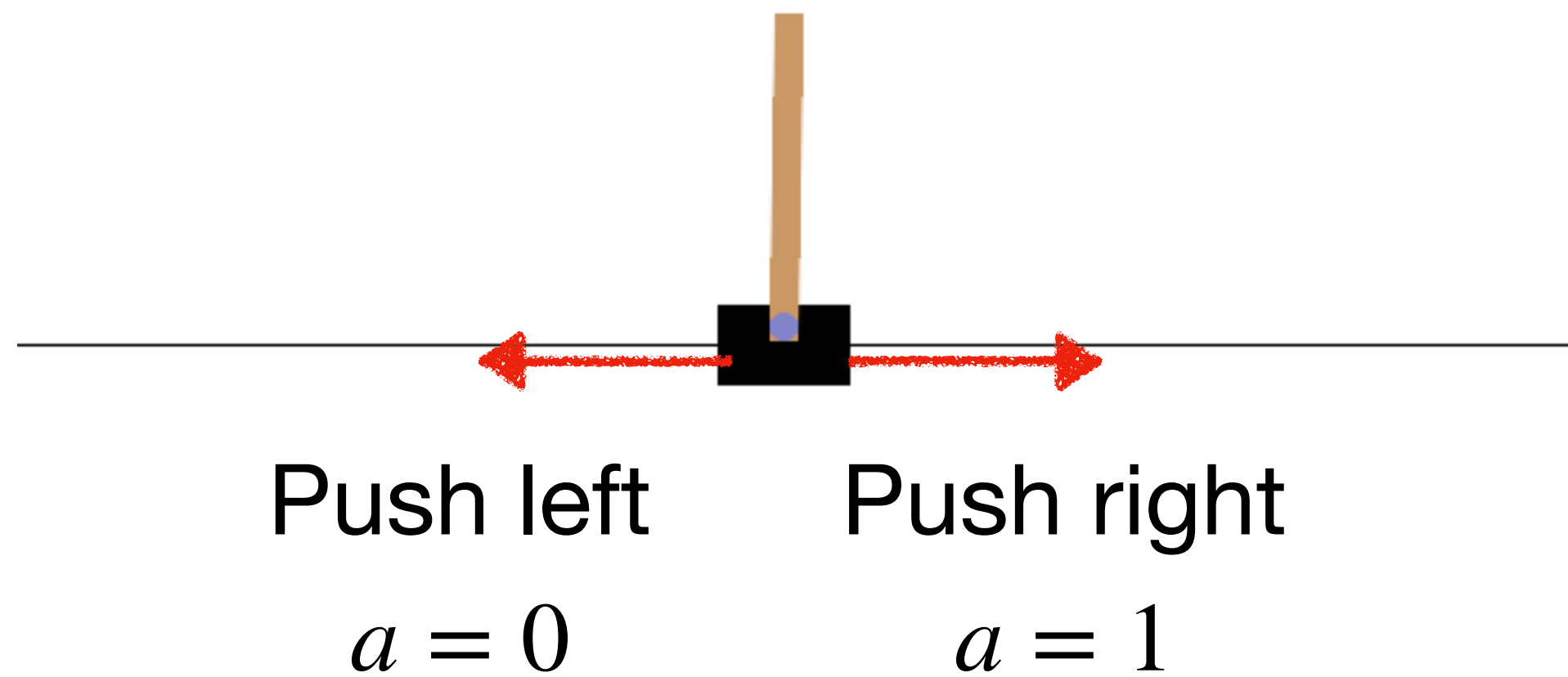
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Policy 2: adaptive

$$\pi(s) = \begin{cases} 0(\text{left}) & \text{if pole angle} < 0 \\ 1(\text{right}) & \text{else} \end{cases}$$

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Outlines:

- ✓ 1. Definitions of Markov Decision Process
- 2. Value functions (V and Q functions)
- 3. Bellman equations

Performance of a policy

Expected total reward of a policy π :

$$V^\pi(s) = \mathbb{E} \left[r_0 + \gamma r_1 + \gamma^2 r_2 + \dots + \gamma^h r_h + \dots \mid s_0 = s, \pi \right]$$

$\gamma \in [0,1)$: discount factor (value future reward less and less)

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Q: think about the CartPole example, is there a way we can estimate $V^\pi(s)$ at a given s ?

Optimal policy

$\pi^\star$: the policy that maximizes expected future reward at all states

$$V^\star(s) \geq V^\pi(s), \forall s, \forall \pi$$

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Q: what is the optimal policy when $\gamma = 0$?

State-action Q function

$$Q^{\pi}(s, a) = \mathbb{E} \left[r_0 + \gamma r_1 + \gamma^2 r_2 + \dots + \gamma^h r_h + \dots \mid s_0 = s, a_0 = a, \pi \right]$$

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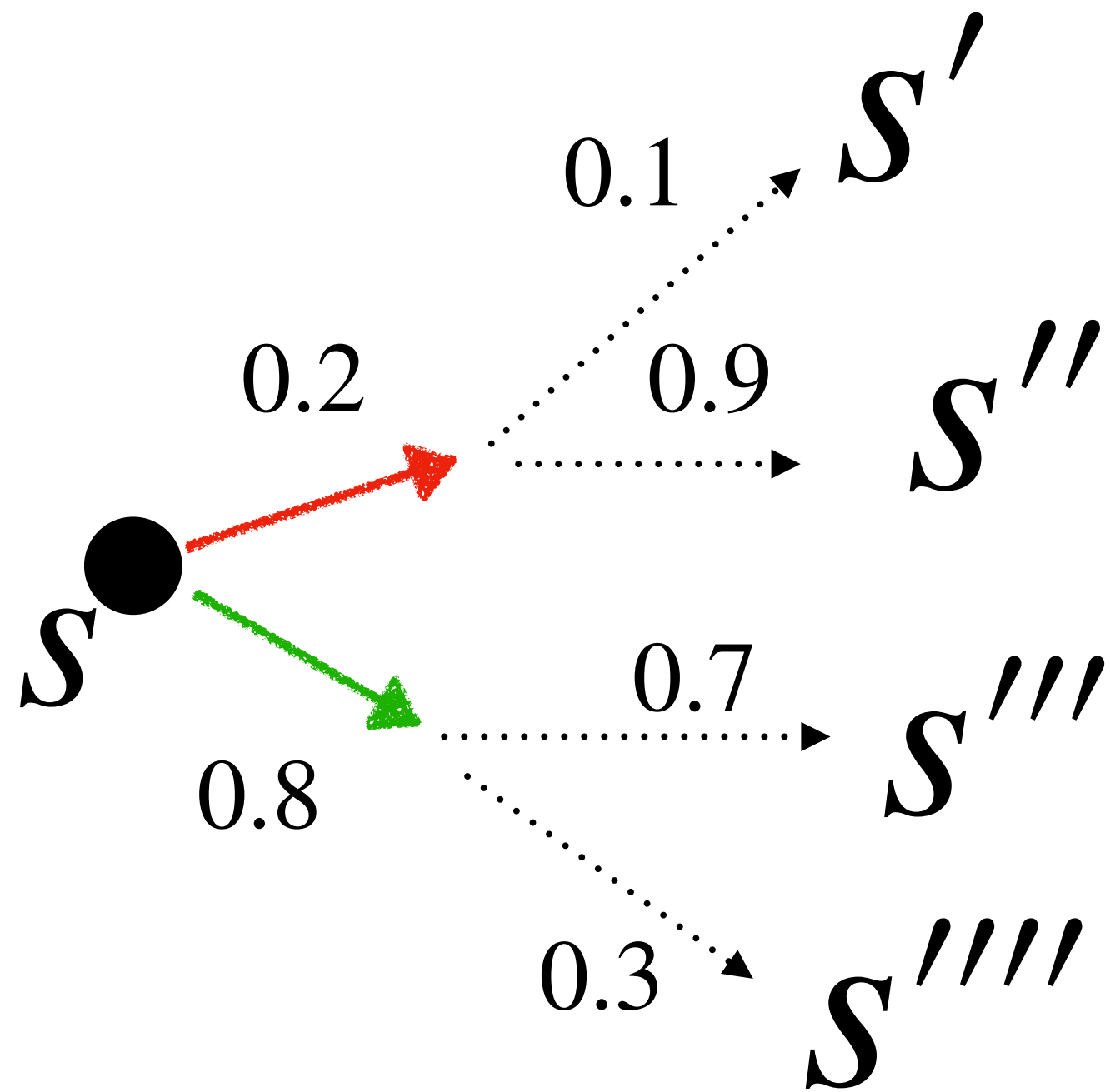
3. Bellman equations

Can we express the value function using only one step?

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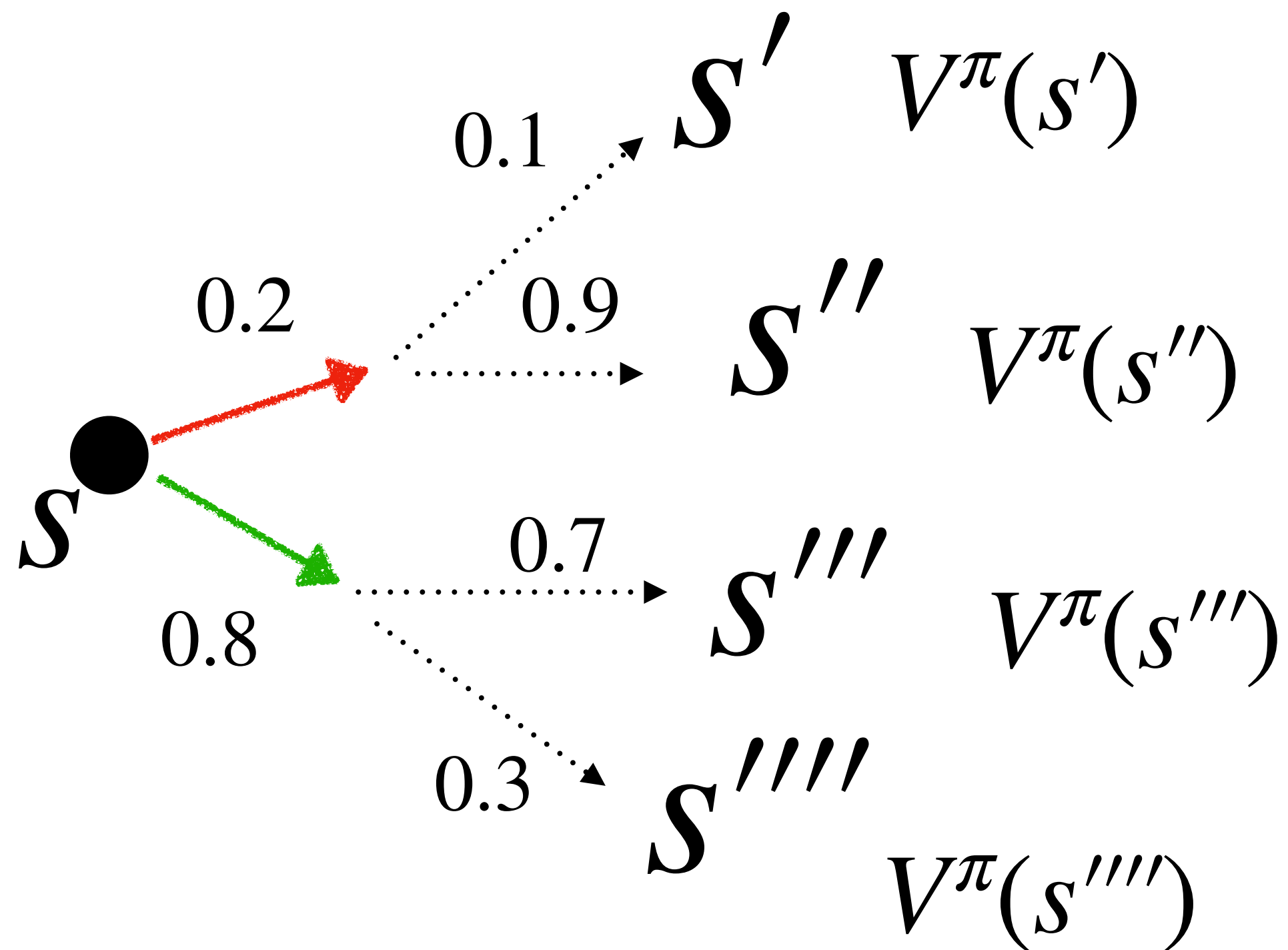
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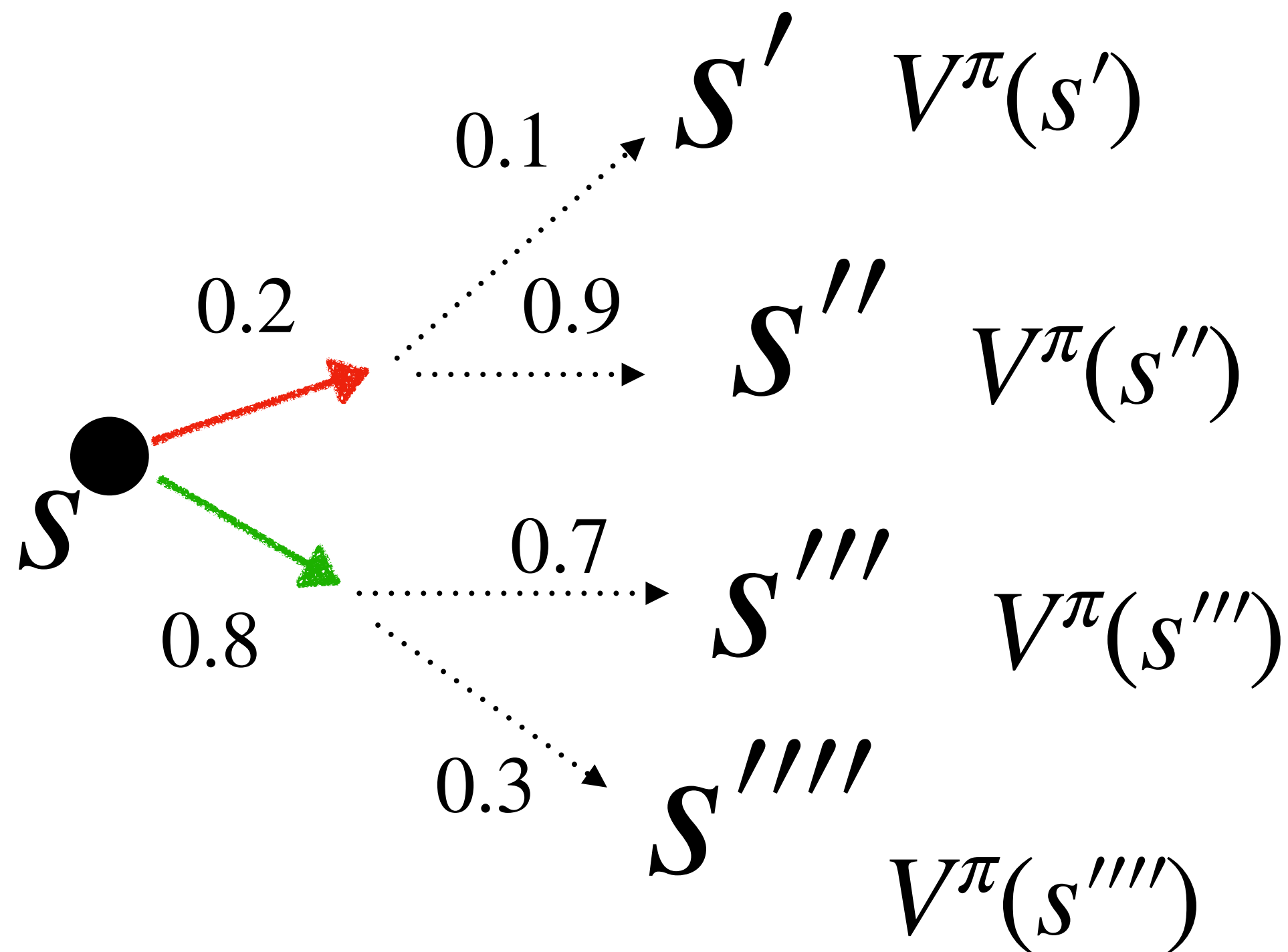
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$$V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)} \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V^\pi(s')] \right]$$

Bellman equation for value function

Can we express the value function using only one step?

Your homework: understand the one-step relationship between V and Q

$$Q^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V^\pi(s')]$$

$$V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot | s)} Q^\pi(s, a)$$

Summary:

Discounted infinite horizon MDP:

State, action, policy, transition, reward (or cost), discount factor

V function and Q function

Key concept: **Bellman equation**

Next Lecture

- Exact Policy Evaluation
- Fixpoint solution
- Approximate Policy Evaluation via an Iterative Algorithm